

ST117 Written Report - Phase 1 (EDA)

Report Pod MS-05

11-05-2026

- Please replace XY-ab in filename.
- Please replace XY-ab, and 1999-12-31 in the header.
- Please read the document “WR Assessment, Style, and Submission Guidance” (see Moodle).
- By submitting this assignment, you confirm that you have read and will adhere to the guidance on AI set out in that document.
- Remember that you are responsible for everything that you submit.
- Please adhere to Word, page, figure, and table limits.
- Note that 20% of the total credit is dedicated to style.
- Please include R code in both Rmd and pdf files.

Contributions

This submission was created by (extend as needed): 1. WARWICK ID:u5603360 1,2,5,6 2. WARWICK ID:u5734835 3,4

load all datasets with same names so it all works as 1 and set working directory to file location

```
setwd(".")
FineGrain <- read.csv("C:/Dev/year 1/Stats Modeling/Final report/Data/FineGrain.csv")
head(FineGrain)
```

##	SITECODE	SYEAR	PLOTPID	CELLID	FIELDNAME	VALUE
## 1	T01	1994	169	4	VEG_SPEC	796
## 2	T01	1994	169	4	VEG_SPEC	990
## 3	T01	1994	169	4	VEG_SPEC	2982
## 4	T01	1994	169	5	VEG_SPEC	122
## 5	T01	1994	169	5	VEG_SPEC	796
## 6	T01	1994	169	5	VEG_SPEC	981

```
FineGrain2 <- read.csv("Data/ECN_VF2.csv")
Baseline<- read.csv("C:/Dev/year 1/Stats Modeling/Final report/Data/Baseline.csv")
head(Baseline)
```

##	SITECODE	SYEAR	PLOTPID	PLOTTYPE	FIELDNAME	VALUE
## 1	T01	1993	7	S	VEG_SPEC	118
## 2	T01	1993	7	S	VEG_SPEC	197
## 3	T01	1993	7	S	VEG_SPEC	222
## 4	T01	1993	7	S	VEG_SPEC	262
## 5	T01	1993	7	S	VEG_SPEC	421
## 6	T01	1993	7	S	VEG_SPEC	433

```
Woodland <- read.csv("C:/Dev/year 1/Stats Modeling/Final report/Data/Woodland.csv")
head(Woodland)
```

##	SITECODE	SYEAR	PLOTPID	CELLID	TREEID	STEMID	TREE_SPEC	FIELDNAME	VALUE
## 1	T05	1994	325	1	1	1	237	DIAMETER	19
## 2	T05	1994	325	1	1	1	237	HEIGHT	25

```
## 3      T05  1994    325      1      1      1      237 NUM_STEMS      1
## 4      T05  1994    325      2      2      1      237  DIAMETER     16
## 5      T05  1994    325      2      2      1      237   HEIGHT     22
## 6      T05  1994    325      2      2      1      237 NUM_STEMS      1

Butterflies<- read.csv("C:/Dev/year 1/Stats Modeling/Final report/Data/Butterflies.csv")
head(Butterflies)
```

```
##  SITECODE LCODE      SDATE SECTION FIELDNAME VALUE BROODED
## 1      T01      1 10-APR-93      5      84      1
## 2      T01      1 10-APR-93      6      XX      0
## 3      T01      1 10-APR-93      7      XX      0
## 4      T01      1 10-APR-93      8      XX      0
## 5      T01      1 10-APR-93      9      2      2
## 6      T01      1 10-APR-93      4      84      2
```

```
Moths <- read.csv("C:/Dev/year 1/Stats Modeling/Final report/Data/Moths.csv")
head(Moths)
```

```
##  SITECODE LCODE      SDATE FIELDNAME VALUE
## 1      T06      2 24-FEB-92      2542      1
## 2      T06      2 24-FEB-92      929      1
## 3      T06      2 25-FEB-92      2542      1
## 4      T06      2 25-FEB-92      929      1
## 5      T06      2 05-MAR-92      907      1
## 6      T06      2 09-MAR-92      571      1
```

Task 1: Context and study design

Limit for all dataset combined: 0.5 pages text in 11pt font with single line spacing.

**The ECN datasets used in this project combine long-term monitoring of vegetation and invertebrates across terrestrial sites in the UK. Although all five datasets are intended to detect ecological change, they differ in what they measure and in the scale at which they operate. The vegetation datasets focus on plant community composition and structure, while the butterfly and moth datasets record animal abundance through repeated counts. Baseline vegetation (VB) provides an initial site-wide characterisation using 2 m × 2 m plots on a regular grid with additional infill points where needed. Fine-grain vegetation (VF) records species presence within randomly selected 40 cm × 40 cm cells inside 10 m × 10 m plots, giving much finer spatial detail. Woodland vegetation (VW) is more specialised and records tree and shrub species, dominance, seedling counts, height and diameter at breast height within 10 m × 10 m woodland plots. The datasets also differ in sampling frequency. VB is mainly a baseline survey, VF is usually repeated every three years, and VW is sampled less often because it is designed to monitor slower structural change in woodland. In contrast, the invertebrate datasets are based on regular repeated counts. Butterflies (IB) are surveyed weekly along fixed transects divided into sections between April and September, while moths (IM) are recorded using standard light traps, usually on a nightly basis. These differences in timing reflect the ecological processes being measured, since vegetation generally changes more slowly than invertebrate abundance and activity. The datasets are structured around different sampling units. Vegetation data are organised by site, plot and sometimes cell, butterfly data by site, transect location and section, and moth data by site, trap location and date. These structural differences mean that each dataset supports different ecological questions. The vegetation datasets are useful for studying community composition, habitat structure and spatial heterogeneity, whereas the invertebrate datasets are better suited to analysing abundance patterns, seasonality and temporal variation. Together, they provide a broader picture of ecological change across sites and years than any one dataset alone.

Task 2: Diagrams

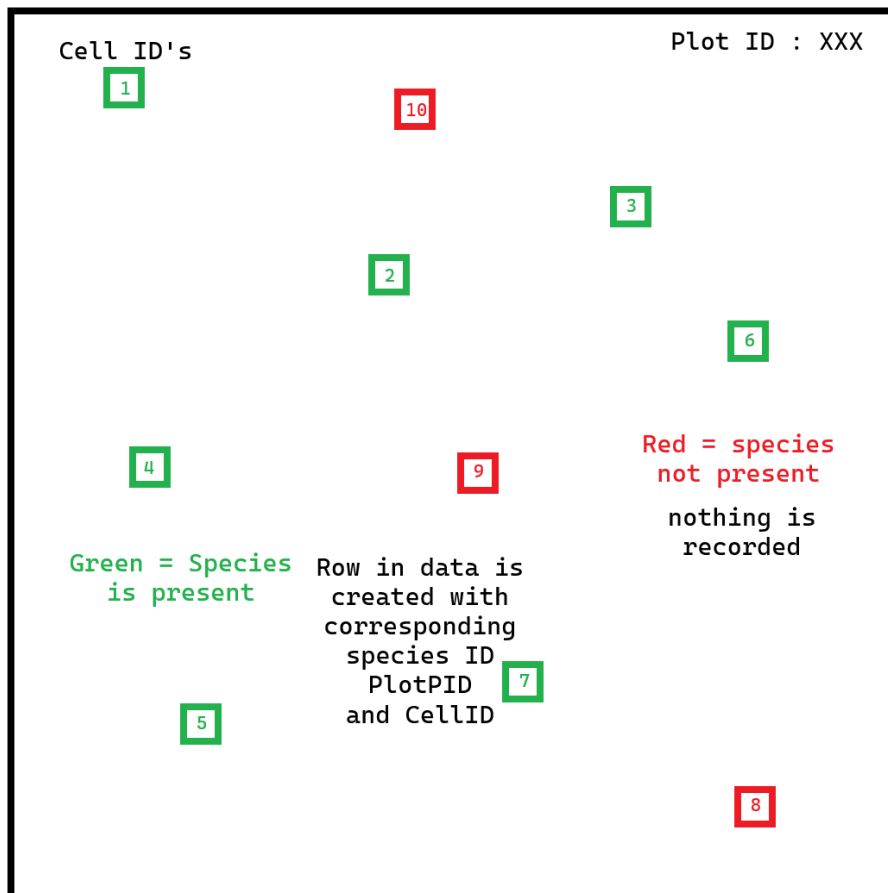
The diagrams can be drawn by hand or created with suitable software.

Task 2: Study Design Diagrams

Fine Grain vegetation

It has been thought more efficient statistically to use a relatively large number of small plots rather than a small number of larger plots. and it is easier to quantify presence of species as opposed to estimated cover and other more subjective and hard to measure phenoms. they use a 40 cm x 40 cm and Every 3 years sample from 10 random cells in the 10m by 10m plotID location randomly selected

```
knitr::include_graphics("C:/Dev/year 1/Stats Modeling/Final report/Images/FineGrainDC.png")
```



Note also that when going to these places analysis is done of both the plot and records of how the individual cells may encounter sample biases . in the form of additional documents. ## Baseline vegetation

Brief description

Baseline vegetation (VB) is a one-off baseline survey. Sampling is based on an approximately regular grid, with additional infill points where needed. The core spatial unit is a 2 m × 2 m plot.

```
plot.new()
plot.window(xlim = c(0, 10), ylim = c(0, 6))
title(main = "VB: baseline vegetation", font.main = 2, cex.main = 1.4)

# Time
text(0.6, 5.0, "Time", adj = 0, font = 2)
arrows(1.2, 4.9, 3.8, 4.9, length = 0.08, lwd = 2)
text(2.2, 5.25, "one-off survey", cex = 0.9)

# Space
text(0.6, 4.0, "Space", adj = 0, font = 2)
rect(1.0, 1.6, 4.1, 3.6, lwd = 2)
for(i in 1:4) lines(c(1.0 + i*0.62, 1.0 + i*0.62), c(1.6, 3.6))
for(j in 1:2) lines(c(1.0, 4.1), c(1.6 + j*0.67, 1.6 + j*0.67))
```

```

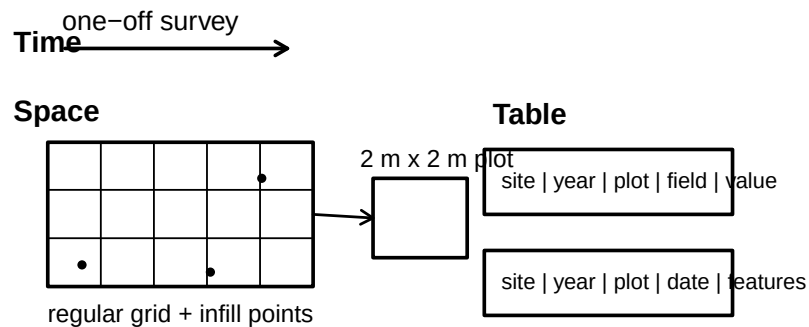
points(c(1.4, 3.5, 2.9), c(1.9, 3.1, 1.8), pch = 16, cex = 0.7)
text(1.0, 1.2, "regular grid + infill points", adj = 0, cex = 0.8)

rect(4.8, 2.0, 5.9, 3.1, lwd = 2)
text(4.65, 3.35, "2 m x 2 m plot", adj = 0, cex = 0.8)
arrows(4.1, 2.6, 4.8, 2.55, length = 0.08, lwd = 1.5)

# Table
text(6.2, 4.0, "Table", adj = 0, font = 2)
rect(6.1, 2.6, 9.0, 3.5, lwd = 2)
text(6.3, 3.05, "site | year | plot | field | value", adj = 0, cex = 0.78)
rect(6.1, 1.2, 9.0, 2.1, lwd = 2)
text(6.3, 1.65, "site | year | plot | date | features", adj = 0, cex = 0.78)

```

VB: baseline vegetation



Woodland Vegetation

Brief description

Woodland vegetation (VW) is used for woodland or scrub plots. It records tree and shrub species, seedlings, height and dbh within 10 m × 10 m woodland plots.

```

plot.new()
plot.window(xlim = c(0, 10), ylim = c(0, 6))
title(main = "VW: woodland vegetation", font.main = 2, cex.main = 1.4)

# Time
text(0.6, 5.0, "Time", adj = 0, font = 2)
arrows(1.2, 4.9, 4.4, 4.9, length = 0.08, lwd = 2)
xs <- c(1.6, 2.9, 4.2)
labs <- c("Y1", "Y10", "Y19")
for(i in 1:3){
  lines(c(xs[i], xs[i]), c(4.8, 5.0), lwd = 2)
  text(xs[i], 5.25, labs[i], cex = 0.85)
}
text(1.9, 4.45, "main survey every 9 years", adj = 0, cex = 0.85)

```

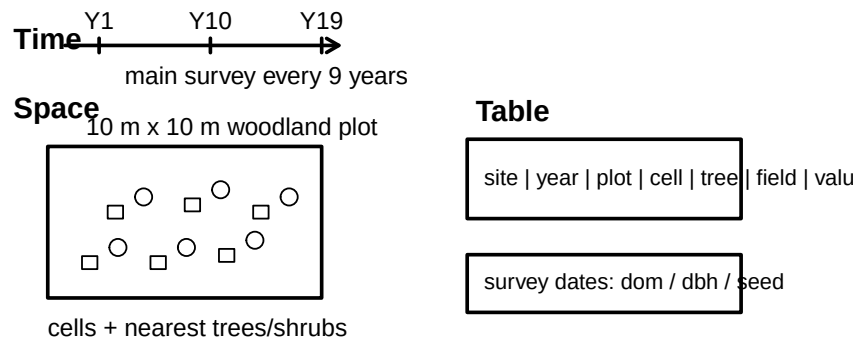
```

# Space
text(0.6, 4.0, "Space", adj = 0, font = 2)
rect(1.0, 1.4, 4.2, 3.5, lwd = 2)
text(1.45, 3.75, "10 m x 10 m woodland plot", adj = 0, cex = 0.8)
pts <- matrix(c(1.4,1.8, 2.2,1.8, 3.0,1.9, 1.7,2.5, 2.6,2.6, 3.4,2.5), ncol = 2, byrow = TRUE)
for(i in 1:nrow(pts)){
  rect(pts[i,1], pts[i,2], pts[i,1]+0.18, pts[i,2]+0.18)
  symbols(pts[i,1]+0.42, pts[i,2]+0.30, circles = 0.10, inches = FALSE, add = TRUE)
}
text(1.0, 1.0, "cells + nearest trees/shrubs", adj = 0, cex = 0.8)

# Table
text(6.0, 4.0, "Table", adj = 0, font = 2)
rect(5.9, 2.5, 9.1, 3.6, lwd = 2)
text(6.1, 3.05, "site | year | plot | cell | tree | field | value", adj = 0, cex = 0.73)
rect(5.9, 1.2, 9.1, 2.0, lwd = 2)
text(6.1, 1.6, "survey dates: dom / dbh / seed", adj = 0, cex = 0.78)

```

VW: woodland vegetation



Butterflies

Brief description

Butterflies (IB) are recorded along fixed transects divided into sections. Surveys are carried out weekly during the butterfly season.

```

plot.new()
plot.window(xlim = c(0, 10), ylim = c(0, 6))
title(main = "IB: butterflies", font.main = 2, cex.main = 1.4)

# Time
text(0.6, 5.0, "Time", adj = 0, font = 2)
arrows(1.2, 4.9, 4.8, 4.9, length = 0.08, lwd = 2)
text(1.7, 5.25, "weekly, Apr-Sep", adj = 0, cex = 0.9)

# Space
text(0.6, 4.0, "Space", adj = 0, font = 2)

```

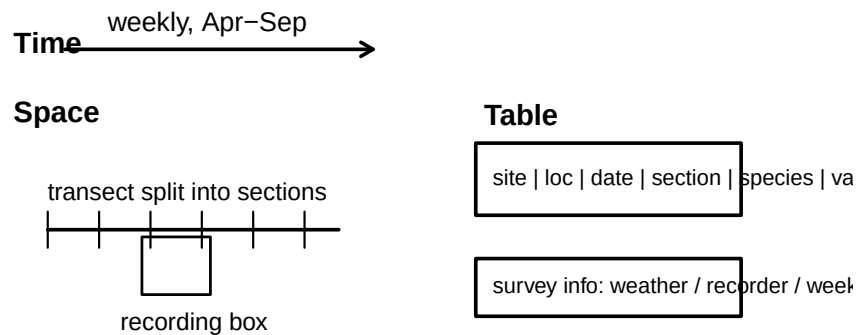
```

lines(c(1.0, 4.4), c(2.4, 2.4), lwd = 2)
for(x in c(1.0, 1.6, 2.2, 2.8, 3.4, 4.0)){
  lines(c(x, x), c(2.15, 2.65), lwd = 1.2)
}
text(1.0, 2.9, "transect split into sections", adj = 0, cex = 0.8)
rect(2.1, 1.5, 2.9, 2.3, lwd = 1.5)
text(1.85, 1.1, "recording box", adj = 0, cex = 0.8)

# Table
text(6.1, 4.0, "Table", adj = 0, font = 2)
rect(6.0, 2.6, 9.1, 3.6, lwd = 2)
text(6.2, 3.1, "site | loc | date | section | species | value", adj = 0, cex = 0.75)
rect(6.0, 1.2, 9.1, 2.0, lwd = 2)
text(6.2, 1.6, "survey info: weather / recorder / week", adj = 0, cex = 0.75)

```

IB: butterflies



Moths

Brief description

Moths (IM) are collected using a standard light trap at fixed locations, usually with nightly sampling.

```

plot.new()
plot.window(xlim = c(0, 10), ylim = c(0, 6))
title(main = "IM: moths", font.main = 2, cex.main = 1.4)

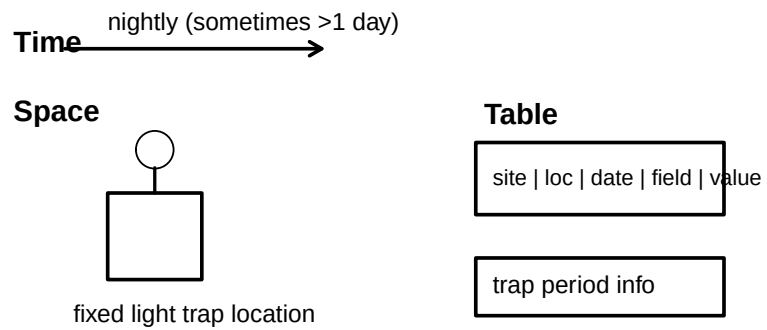
# Time
text(0.6, 5.0, "Time", adj = 0, font = 2)
arrows(1.2, 4.9, 4.2, 4.9, length = 0.08, lwd = 2)
text(1.7, 5.25, "nightly (sometimes >1 day)", adj = 0, cex = 0.85)

# Space
text(0.6, 4.0, "Space", adj = 0, font = 2)
rect(1.7, 1.7, 2.8, 2.9, lwd = 2)
symbols(2.25, 3.5, circles = 0.22, inches = FALSE, add = TRUE)
lines(c(2.25, 2.25), c(2.9, 3.25), lwd = 1.8)
text(1.3, 1.2, "fixed light trap location", adj = 0, cex = 0.8)

```

```
# Table
text(6.1, 4.0, "Table", adj = 0, font = 2)
rect(6.0, 2.6, 8.9, 3.6, lwd = 2)
text(6.2, 3.1, "site | loc | date | field | value", adj = 0, cex = 0.78)
rect(6.0, 1.2, 8.9, 2.0, lwd = 2)
text(6.2, 1.6, "trap period info", adj = 0, cex = 0.8)
```

IM: moths



Task 3: Data quality assessment

Limits per dataset: 0.5 pages text in 11pt font with single line spacing + 2 figures + 1 table.

Fine-grain vegetation

###General overview

```
# to do this we will remove and review quality codes and observe the species per cell per plot
quality <- FineGrain[FineGrain$FIELDNAME != "VEG_SPEC" ,]
```

```
QualityCodes<-ggplot(quality, aes(x = FIELDNAME , fill = SITECODE)) +geom_bar() + labs(title = "Qual
```

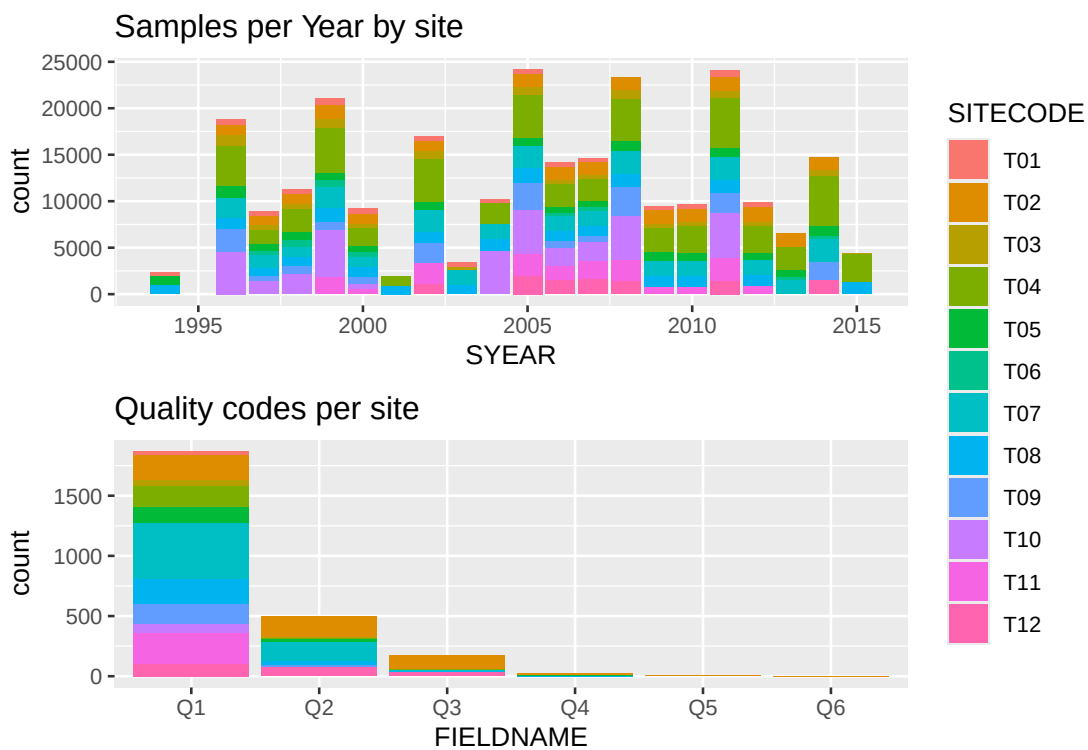
```
# now veiw species data and again stratify by site
```

```
species <- FineGrain[FineGrain$FIELDNAME == "VEG_SPEC" ,]
```

```
DatapointsPerYear<-ggplot(species, aes(x = SYEAR , fill = SITECODE)) +geom_bar() + labs(title = "San
```

```
#plot the plots
```

```
DatapointsPerYear/QualityCodes +plot_layout(guides = "collect")
```



The year by year plots have very strong every 3 year looking tendency in just general numbers . in terms of missingness you cannot tell much from this , however it appears to have a less species recorded in later years or less plots sampled when really you would expect either more data coming in or more sites implemented however there is no available data on the last 10 years and this suggests maybe a reason for the tail off which also questions the completeness of the data.

It doesn't actually specify what the Q's mean in the Documentation, from there being an exponential decay it may be an indicator of the severity of the problem in the sampling. I suspect the Q's describe how it may effect the output of data analysis. and they have values associated with varying reasons of either data censoring or things that may effect the fairness of test eg Sheep Grazing or local boggie stream .

###Timeliness it is very hard to tell if these are accurate measurement of timeliness but the code should contain no logical errors and i have made an individual graph for each site

```
FineGrainArr <- arrange(FineGrain,SITECODE,PLOTPID,SYEAR)
```

```
FineGrain_assesments <- FineGrainArr[FineGrain$FIELDNAME == "VEG_SPEC",]
```

```
# removing the quality codes
```

```
# groups by location
```



```

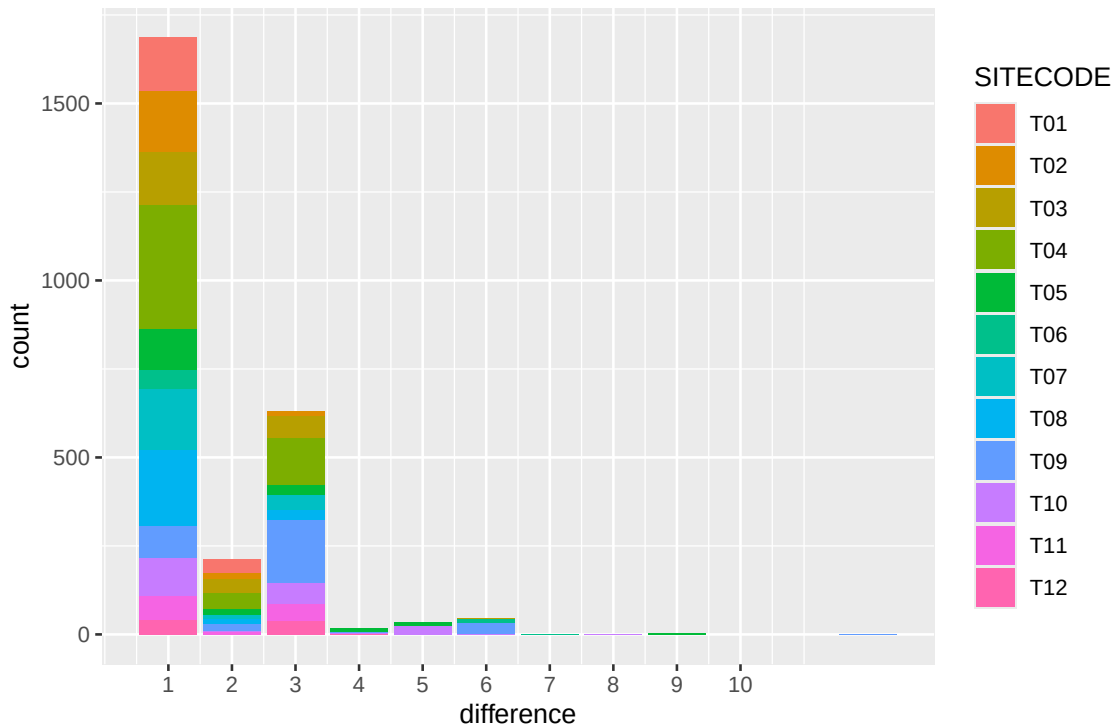
yearPLOTS_FineGrain <- group_by(FineGrain_assesments ,SITECODE , PLOTPID)
# distinctive the rows per group .

TimeStaggers <- mutate(distinct(yearPLOTS_FineGrain,SYEAR),difference = SYEAR-lag(SYEAR))

TimeStaggers_clean <- TimeStaggers[!is.na(TimeStaggers$difference),]
ggplot(TimeStaggers_clean,aes(x = difference, fill = SITECODE))+ geom_bar()+scale_x_continuous(break

```

Difference(years) between Consecutive sample's, Mean difference:1.7406



note that the data suggested a 3 year time difference between times of samples .The data has outputted 1.74 and this is not as expected and may effect how we treat the data. if anything with agencies like this you would expect a lower sample frequency than expected as you need to not overtrample your datasites . this also may be censored as if a datapoint was rechorded two years in a row and never rechorded again and thia happened frequently it would majorly skew the data ## Woodland

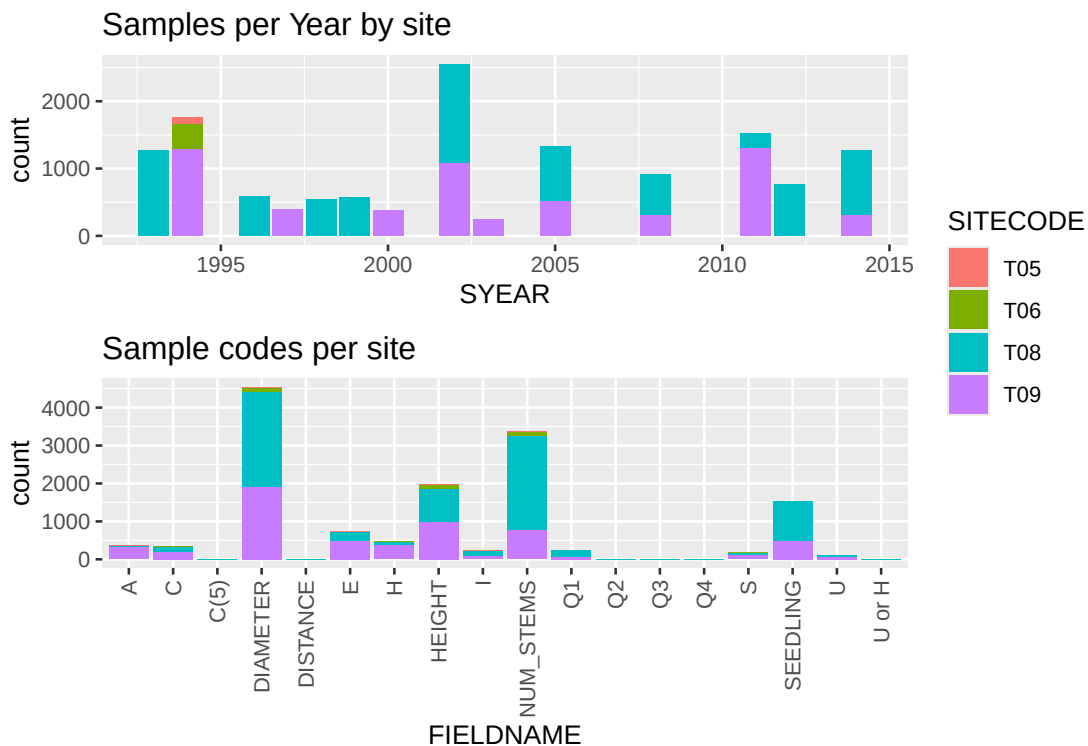
```

# to do this we will remove and review quality codes and observe the species per cell per plot
quality <-Woodland
QualityCodes<-ggplot(quality, aes(x = FIELDNAME , fill = SITECODE)) +geom_bar() + labs(title = "Sampl

# now veiw species data and again stratify by site
species <-Woodland
DatapointsPerYear<-ggplot(species, aes(x = SYEAR , fill = SITECODE)) +geom_bar(position = "stack") +

#plot the plots
DatapointsPerYear/QualityCodes +plot_layout(guides = "collect")

```



note only certain places in sites 5,6,8,9 were sampled from , Samples By year is interesting as T05 and T06 are seemingly only rechecked once which is useless , or after being rechecked the areas have been under controlled forestry and cut or something . no meaningfully obvious 9 year trend present unlike the Fine Grain .

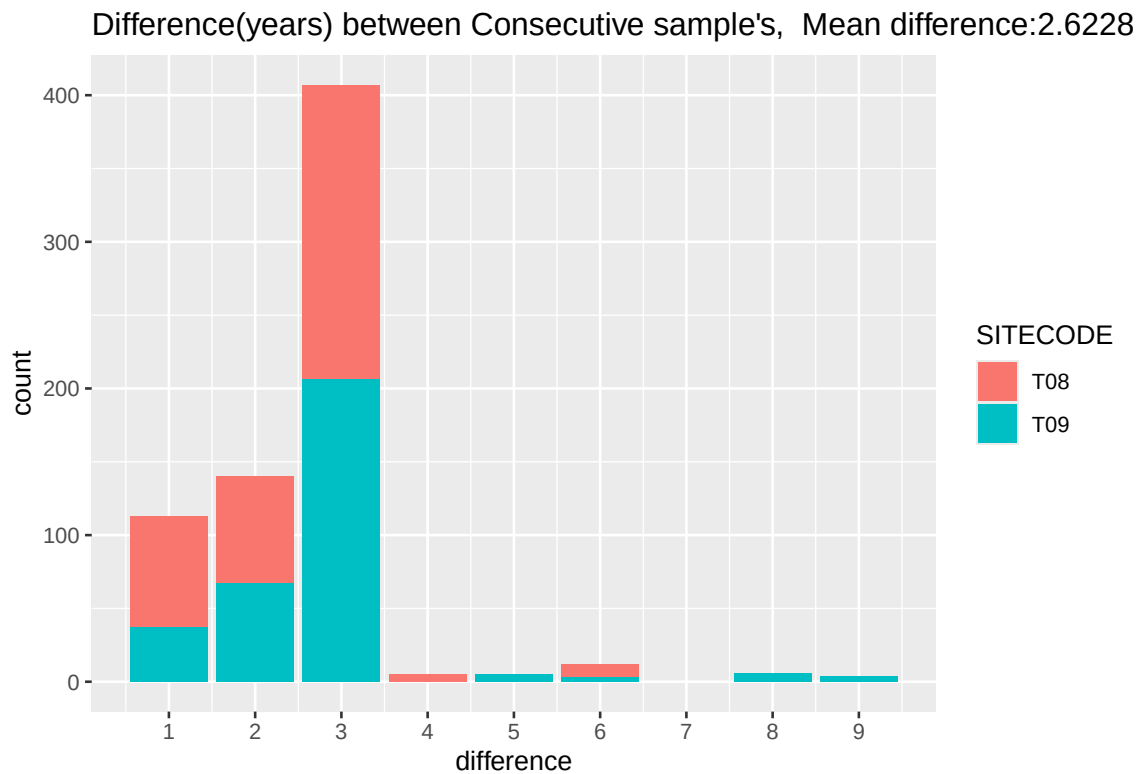
the Sample codes tell us how much of each type of data was recorded and we can see large disparities in what is expected as for each diameter of a tree you would expect a height . this may have been harder to measure possibly suggesting some data needs to be filled in , perhaps by a linear regression

```
WoodlandArr <- arrange(Woodland,SITECODE,PLOTPID,SYEAR)
# no need to remove any datapoints as distinct should do it

# groups by location
yearPLOTS_Woodland <- group_by(WoodlandArr ,SITECODE , PLOTPID)
# distinctive the rows per group .

TimeStaggers <- mutate(distinct(yearPLOTS_Woodland,SYEAR),difference = SYEAR-lag(SYEAR))

TimeStaggers_clean <- TimeStaggers[!is.na(TimeStaggers$difference),]
ggplot(TimeStaggers_clean,aes(x = difference, fill = SITECODE))+ geom_bar()+scale_x_continuous(breaks = 0:10)
```

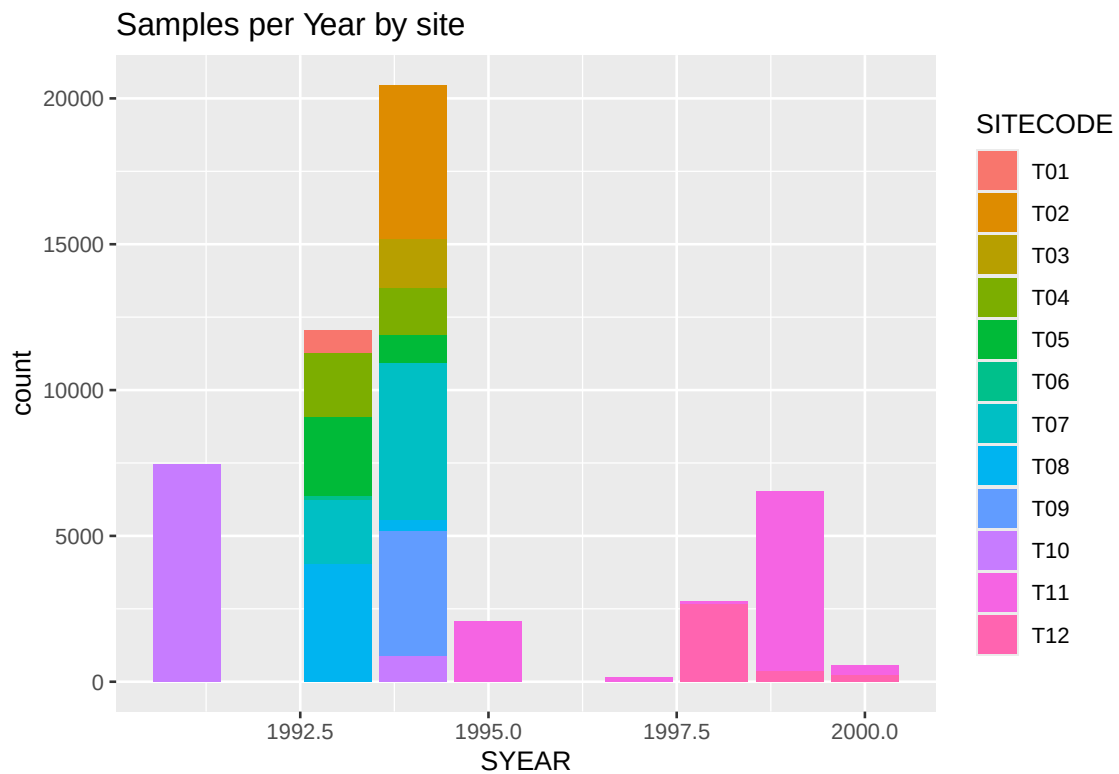


again not as expected as 2.6 « 9 years , i expect some censoring here as well as this is ridiculous and the number of 1 2 and 3 year gaps is very not expected .aswell as the fact there must only be one rechecking of T05 and T06 as there isnt an entry for them as the legend doesnt contain them . ## Baseline vegetation

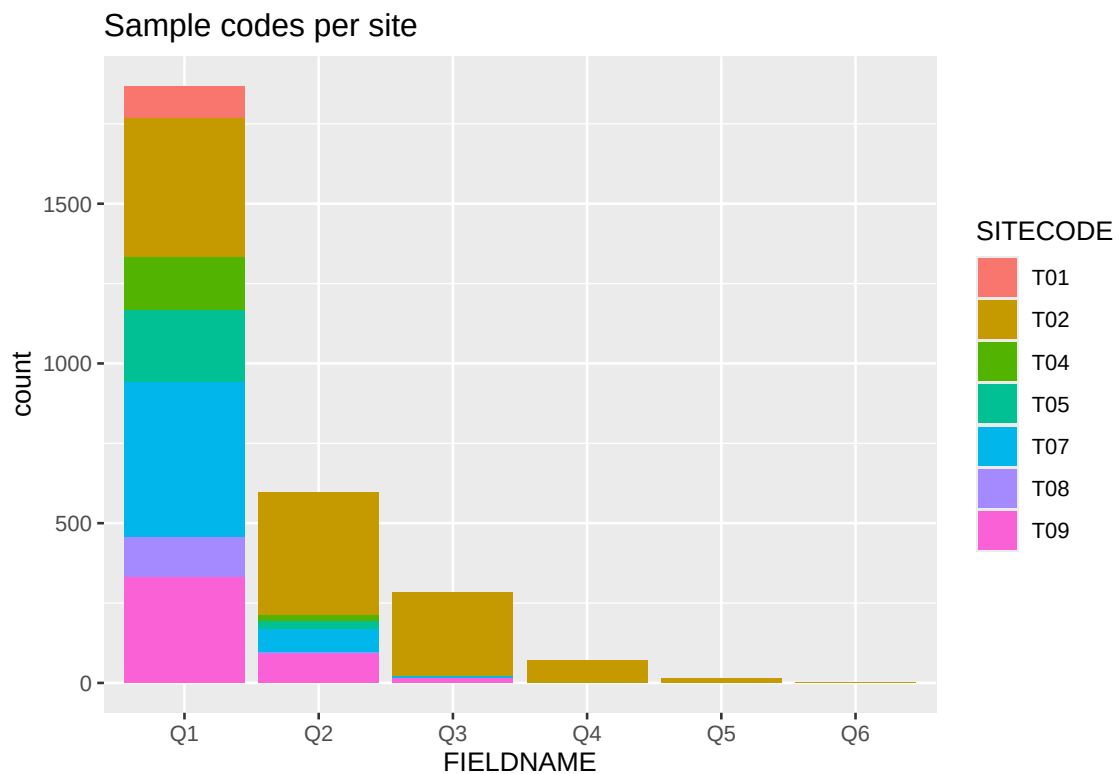
```
# to do this we will remove and review quality codes and observe the species per cell per plot
quality <- Baseline[Baseline$FIELDNAME!= "VEG_SPEC", ]
QualityCodes2<-ggplot(quality, aes(x = FIELDNAME , fill = SITECODE)) +geom_bar() + labs(title = "Sample Quality Codes")

# now view species data and again stratify by site
species <-Baseline
DatapointsPerYear2<-ggplot(species, aes(x = SYEAR , fill = SITECODE)) +geom_bar(position = "stack")

#plot the plots
DatapointsPerYear2
```



QualityCodes2



this data is only sampled once so needs no temporal analysis and you can see it in the Per year plot and the quality code distribution is very similar to others however a very large proportion of T02 is defective and may be enough to consider throwing out T2 from analysis.

Butterflies

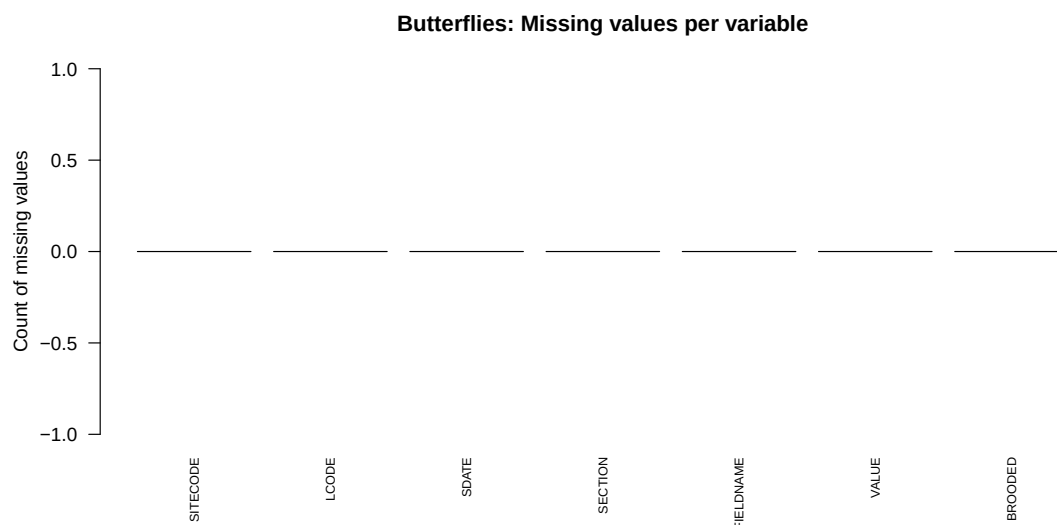
Consistency The data seems to be consistent. Completeness Approximately 13.3% of the data has missing Twice sites have been unable to collect data due to poor weather, both of these were site T02 in 1994SDATES Timeliness Data is hardly collected for T12, it is only collected for a few years

around 2005 and additionally many sites stop collecting data after 2011. Validity There may be an outlier of 234 butterflies spotted in 2006 in T04.

One major issue is that there are a lot of missing SDATES, primarily in site codes T08, T01, T05 and T06. This means that some of the data can't be used to compare changes over time and will cause incomplete data but we don't know where. Additionally most of the entries are XX which corresponds to no butterflies being seen so there is a very high zero inflation which needs to be removed or dealt with, the use of XX instead of 0 is problematic because sometimes you might want to know how many of them are zeros but in this case it needs to be changed to a numerical value to use.

```
bf <- Butterflies
missing_counts <- colSums(is.na(bf))

par(mar = c(10, 4, 4, 2))
barplot(missing_counts,
        col = "orange",
        las = 2,
        cex.names = 0.7,
        main = "Butterflies: Missing values per variable",
        ylab = "Count of missing values")
```



```
missing_counts
```

```
## SITECODE    LCODE    SDATE    SECTION FIELDNAME    VALUE    BROODED
##          0          0          0          0          0          0          0
```

Moths

Consistency There seem to be no duplicates and the data is consistent. Completeness There are no columns that have any NA values however there is no data for T11 so it is hard to compare with other data sets that do have data for T11. Timeliness Site T12 wasn't covered at all before 2001 but aside from that there are very few gaps and sampling was carried out pretty much every year in every site in the dataset. Validity Site T06 in 1996 has a VALUE of 1999 which is much higher than all of the other values. It is clearly an outlier and may have come from a data entry error.

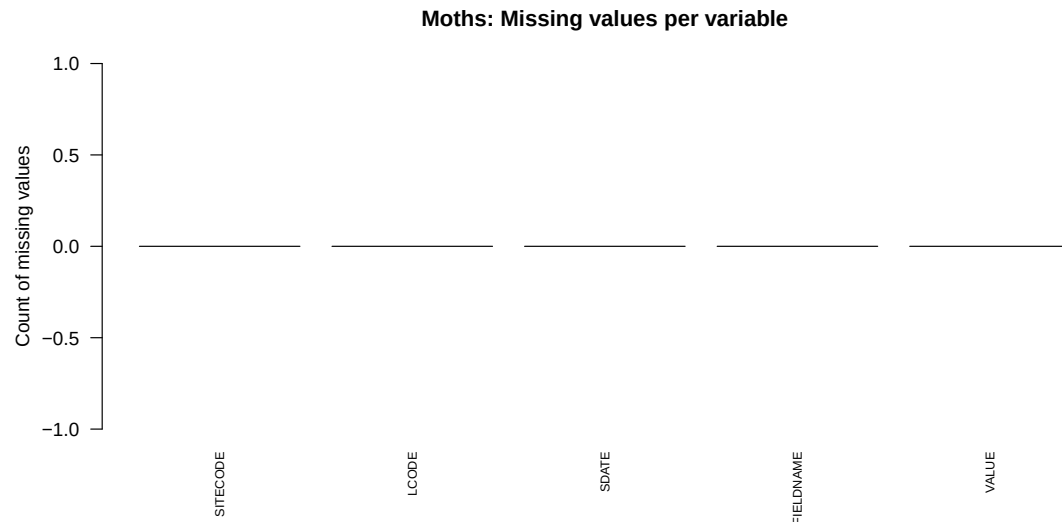
One problem to do with missing data is the fact that T11 is not covered at all, this is problematic because it is a whole site that has not been counted, causing overall totals to decrease significantly and make it incomparable to say the butterflies dataset if you are going to include T11 as the butterflies will always be inflated. Another problem is the lack of data for T12 from before 2001, this makes it harder to compare growth/decrease of number of species, total moths or whatever

else you want to compare timewise, it also means that totals are going to increase after 2000 so you might as well not include T12 at all when looking at changes over time.

```
mt <- Moths

missing_counts <- colSums(is.na(mt))

par(mar = c(10, 4, 4, 2))
barplot(missing_counts,
        col = "purple",
        las = 2,
        cex.names = 0.7,
        main = "Moths: Missing values per variable",
        ylab = "Count of missing values")
```



```
missing_counts

## SITECODE    LCODE    SDATE FIELDNAME    VALUE
##          0         0         0         0         0
```

Task 4: Exploratory data analysis

Limits per dataset: 0.5 pages text in 11pt font with single line spacing + 4 figures + 1 table.

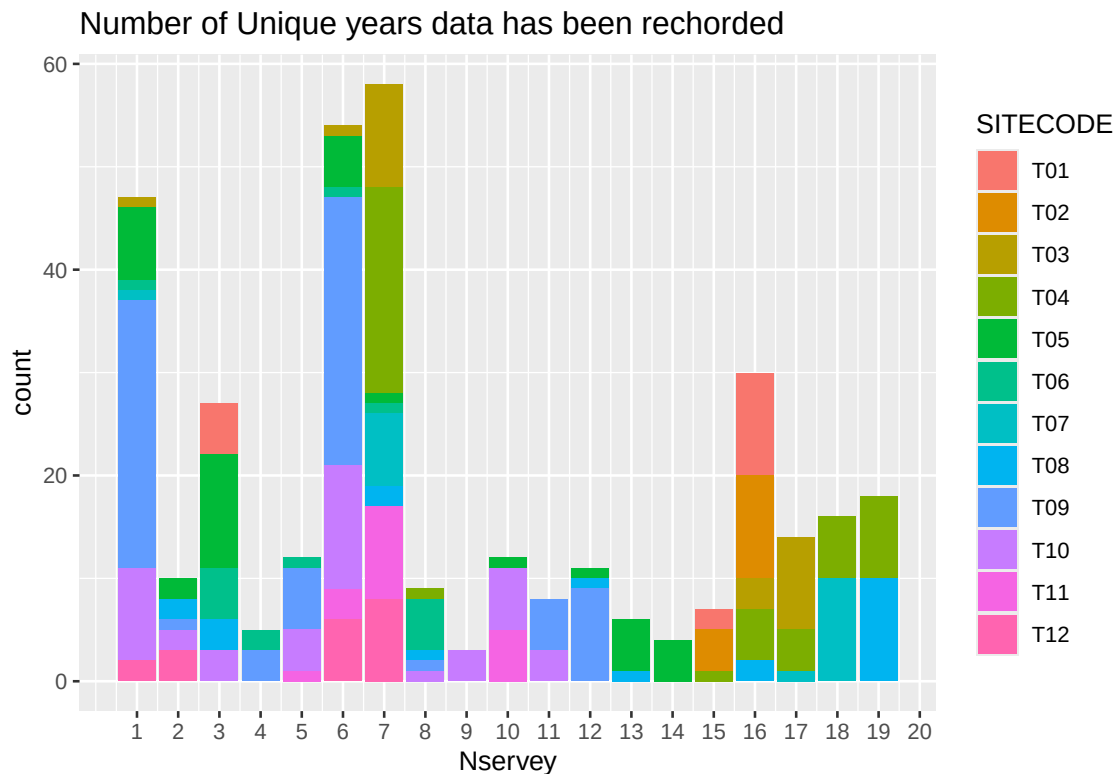
Fine-grain vegetation

```
## how many assesments per plot
yearPLOTS_FineGrain <- group_by(FineGrain , PLOTPID,SITECODE)

APP<-summarise(yearPLOTS_FineGrain,Nservey = n_distinct(SYEAR) , lengtOT = diff(range(SYEAR)))

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by PLOTPID and SITECODE.
## i Output is grouped by PLOTPID.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(PLOTPID, SITECODE))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
ggplot(APP,aes(x =Nsurvey ,fill = SITECODE))+ geom_bar()+scale_x_continuous(breaks = 1:25) +labs(tit
```



```
# 245 plots all of them measured approx over 20 years as some start late and some are weird
mean(APP$Nsurvey)
```

```
## [1] 8.501425
```

```
mean(APP$lengthOT)
```

```
## [1] 13.05698
```

```
mean(APP$lengthOT)/mean(APP$Nsurvey)
```

```
## [1] 1.535858
```

```
# looking for some time dependent features of the graph
```

```
yearPLOTS_FineGrain <- group_by(FineGrain , SYEAR , PLOTPID,VALUE)
```

```
plantindex =summarise(yearPLOTS_FineGrain,NcoveredPlots =n()/10) %>% summarise( plotmean = mean(Ncov
```

```
## `summarise()` has regrouped the output.
```

```
## `summarise()` has regrouped the output.
```

```
## i Summaries were computed grouped by SYEAR, PLOTPID, and VALUE.
```

```
## i Output is grouped by SYEAR and PLOTPID.
```

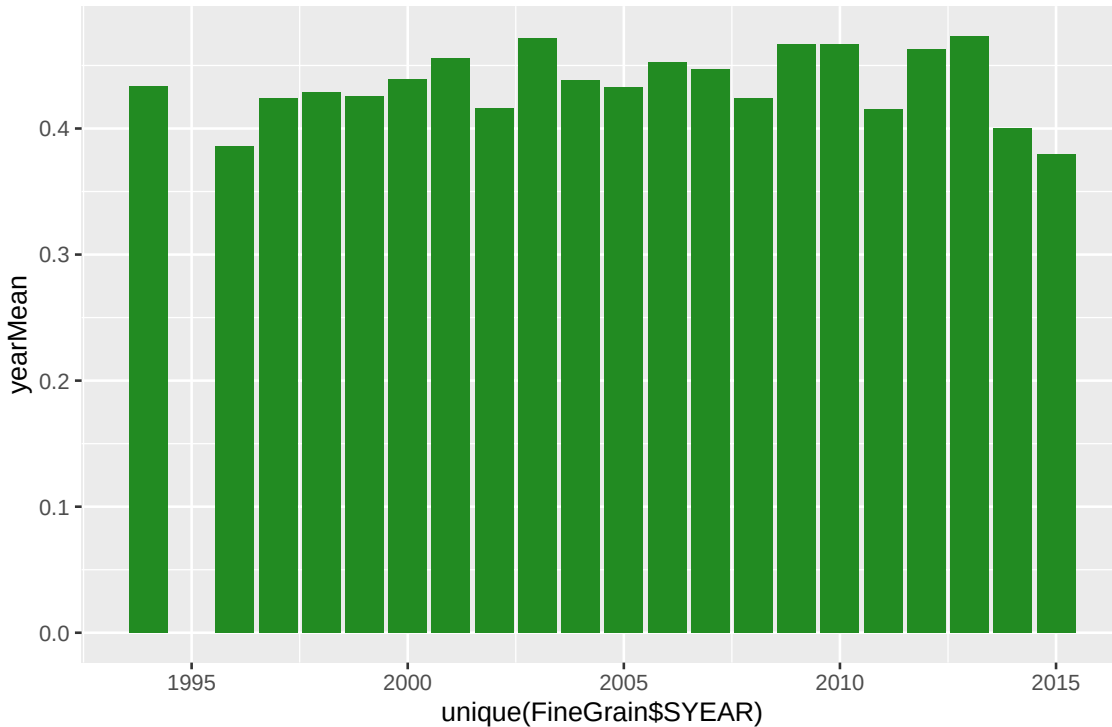
```
## i Use `summarise(.groups = "drop_last")` to silence this message.
```

```
## i Use `summarise(.by = c(SYEAR, PLOTPID, VALUE))` for per-operation grouping
```

```
## (`?dplyr::dplyr_by`) instead.
```

```
ggplot(plantindex, aes(x =unique(FineGrain$SYEAR) , y = yearMean))+geom_col(fill = "forestgreen") +
```

average percentage coverage of a plant over a Plot

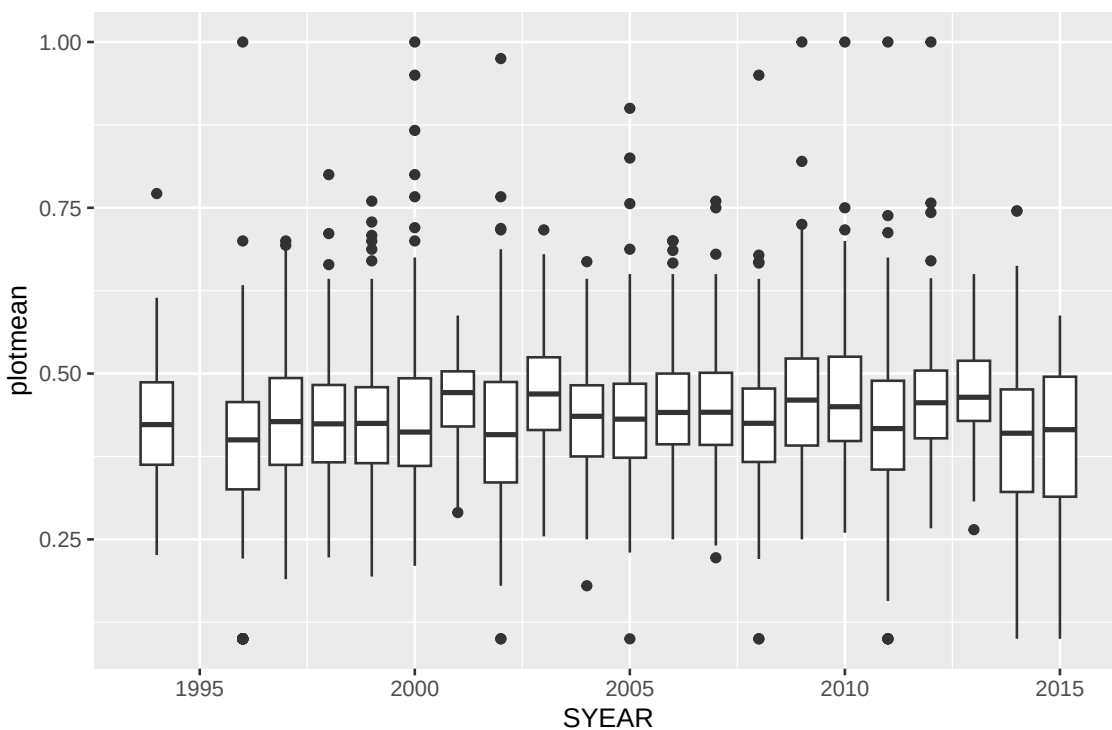


```
plantBoxes = summarise(yearPLOTS_FineGrain, NcoveredPlots = n()/10) %>% summarise( plotmean = mean(NcoveredPlots))

## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by SYEAR, PLOTPID, and VALUE.
## i Output is grouped by SYEAR and PLOTPID.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(SYEAR, PLOTPID, VALUE))` for per-operation grouping
## (`?dplyr::dplyr_by`) instead.

ggplot(plantBoxes, aes(x = SYEAR, group = SYEAR, y = plotmean)) + geom_boxplot() + labs(title = "Boxplot of coverage Per plant over a Plot")
```

Boxplots of coverage Per plant over a Plot



this shows the Number of surveys per site over the 22 years these sites have been running for , we calculated another average for timeliness at 1.54 to back the claims of the earlier calculation of it being bizarrely low however the modal values of 6 and 7 are as expected . note this should be a much less censored version of the last way we calculated this and it gives a lower estimate

for the percentage coverage the box plots show a kind of decline in the later years of sampling but this could be because fewer samples were recorded at these times but the skew is definitely more downwards. but generally this doesnt show much in the way of the fact a majority of plants if there present they tend to be covering a larger amount of the plot .

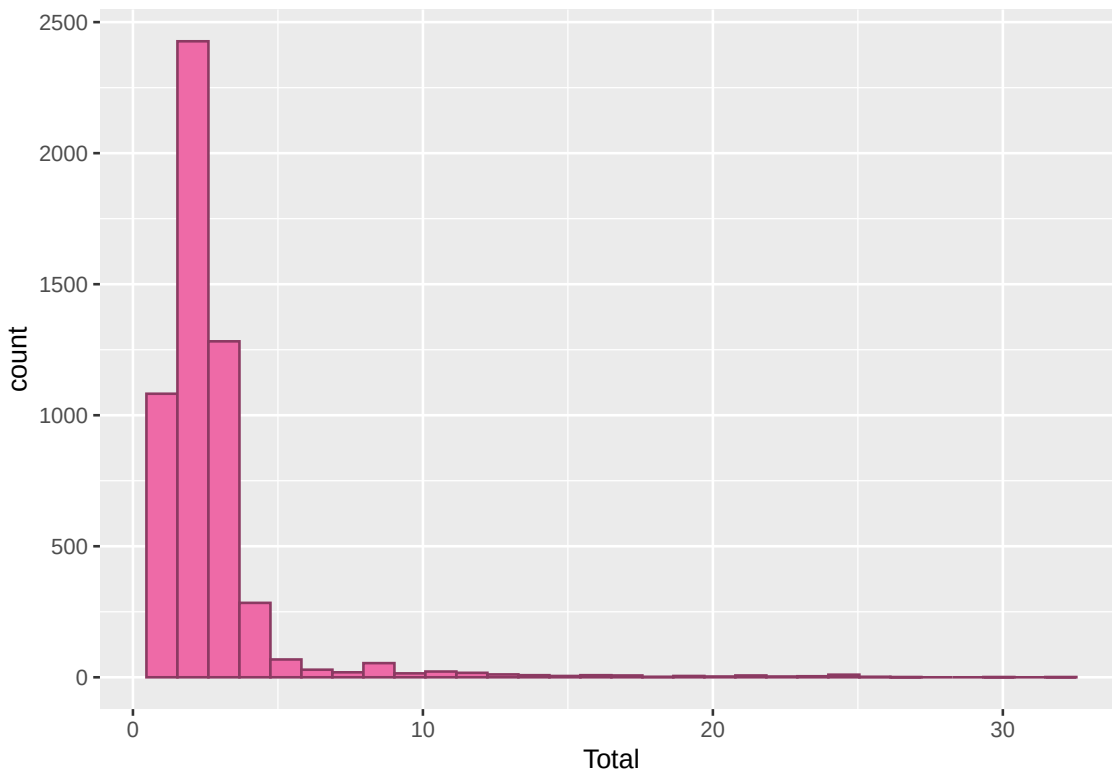
Woodland

```
#the distribution of total vegetation count in each sample
woodland2 = Woodland%>%
  group_by(SITECODE,SYEAR,PLOTPID,CELLID)%>%
  summarise(Total=n())
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by SITECODE, SYEAR, PLOTPID, and CELLID.
## i Output is grouped by SITECODE, SYEAR, and PLOTPID.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(SITECODE, SYEAR, PLOTPID, CELLID))` for per-operation
## grouping (`?dplyr::dplyr_by`) instead.
```

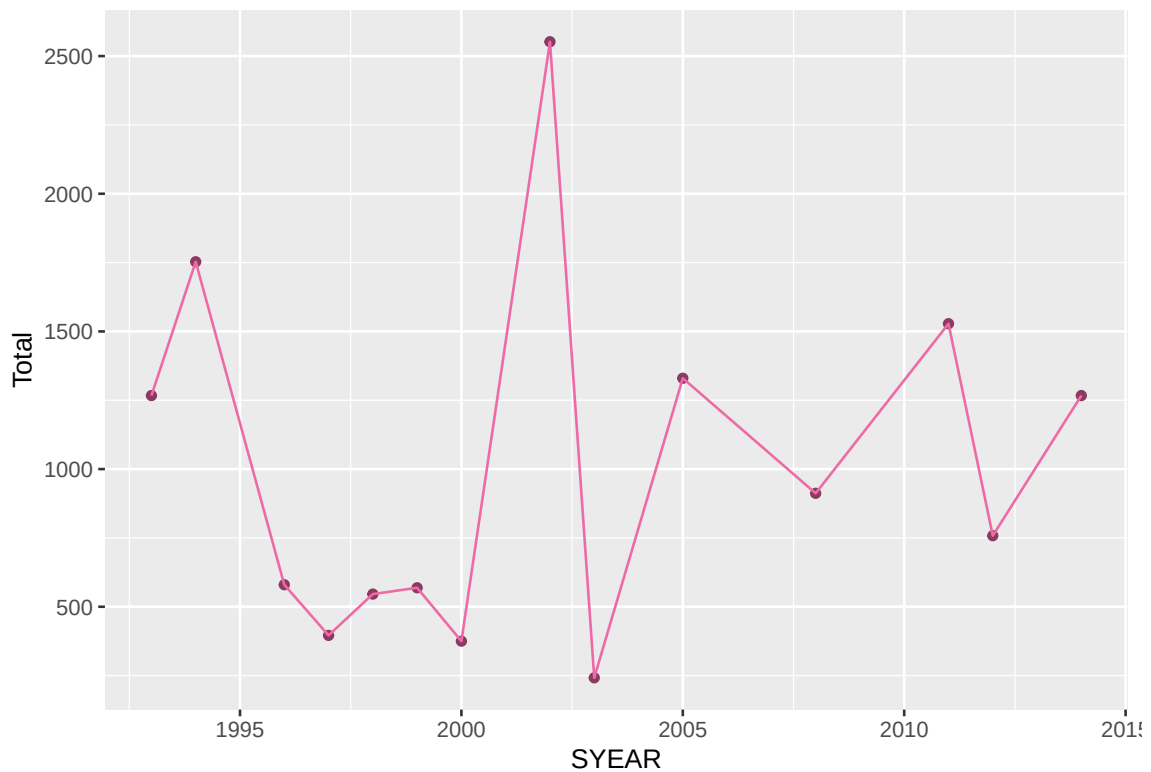
```
ggplot(woodland2,aes(Total))+
  geom_histogram(col="hotpink4",fill="hotpink2")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
#variation over time
woodland3=Woodland%>%
  group_by(SYEAR)%>%
  summarise(Total=n(),na.rm=TRUE)

ggplot(woodland3,aes(SYEAR,Total))+
  geom_point(col="hotpink4")+
  geom_line(col="hotpink2")
```



#spatial variation

```
woodland <- Woodland%>%
  group_by(SYEAR,SITECODE)%>%
  summarise(Total=(n()),na.rm=TRUE)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by SYEAR and SITECODE.
## i Output is grouped by SYEAR.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(SYEAR, SITECODE))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

#stacked bar chart

```
ggplot(woodland,aes(SYEAR>Total,fill=SITECODE))+
  geom_bar(stat="identity")
```



a. The variable chosen is total vegetation per sample, there were less samples of woodland vegetation than finegrain and they also resulted in fewer things being reported, between 0 and 25. The skew is much more significant in this case, with the mode being 2, almost all of the data is less than 5 and there are some outliers that go past that. Again, there are no zeros because nothing was reported so there was nothing to put in the data set.

b. We investigate total number of things found in all of the samples each year. I use a line graph to show how the pattern changes as time passes. It shows inconsistency in the totals, this may be because of inconsistent sampling, samples not being taken when they were supposed to so not all sites' numbers are actually accounted for.

c. We investigate number of species reported in each site each year, i use a stacked bar chart to show each site's number as a proportion of the total. It shows that sampling is inconsistent across sites, for example T05 and T06 only have results for one year so either they found nothing or they were hardly sampled. This likely is the cause of the results found in b.

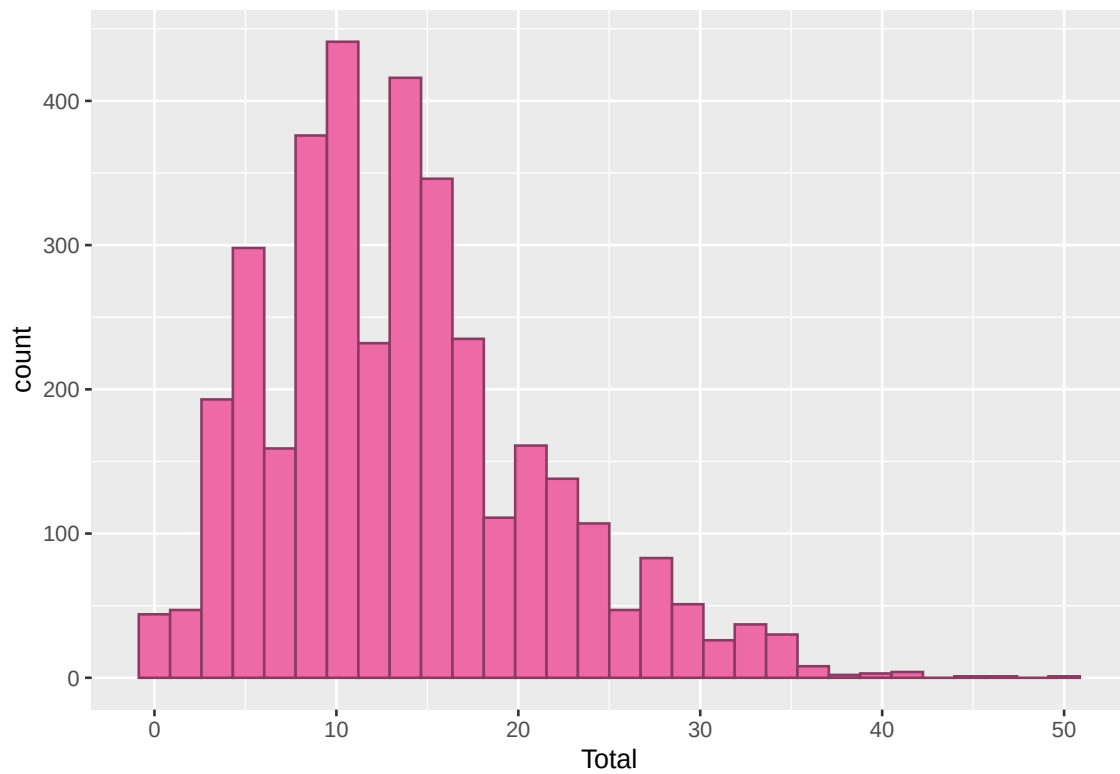
Baseline vegetation

```
#the distribution of total vegetation count in each sample
baseline3 = Baseline%>%
  group_by(SITECODE,SYEAR,PLOTPID)%>%
  summarise(Total=(n_distinct(VALUE[FIELDNAME=="VEG_SPEC"]),na.rm=TRUE)))

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by SITECODE, SYEAR, and PLOTPID.
## i Output is grouped by SITECODE and SYEAR.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(SITECODE, SYEAR, PLOTPID))` for per-operation grouping
##   (?dplyr::dplyr_by`) instead.

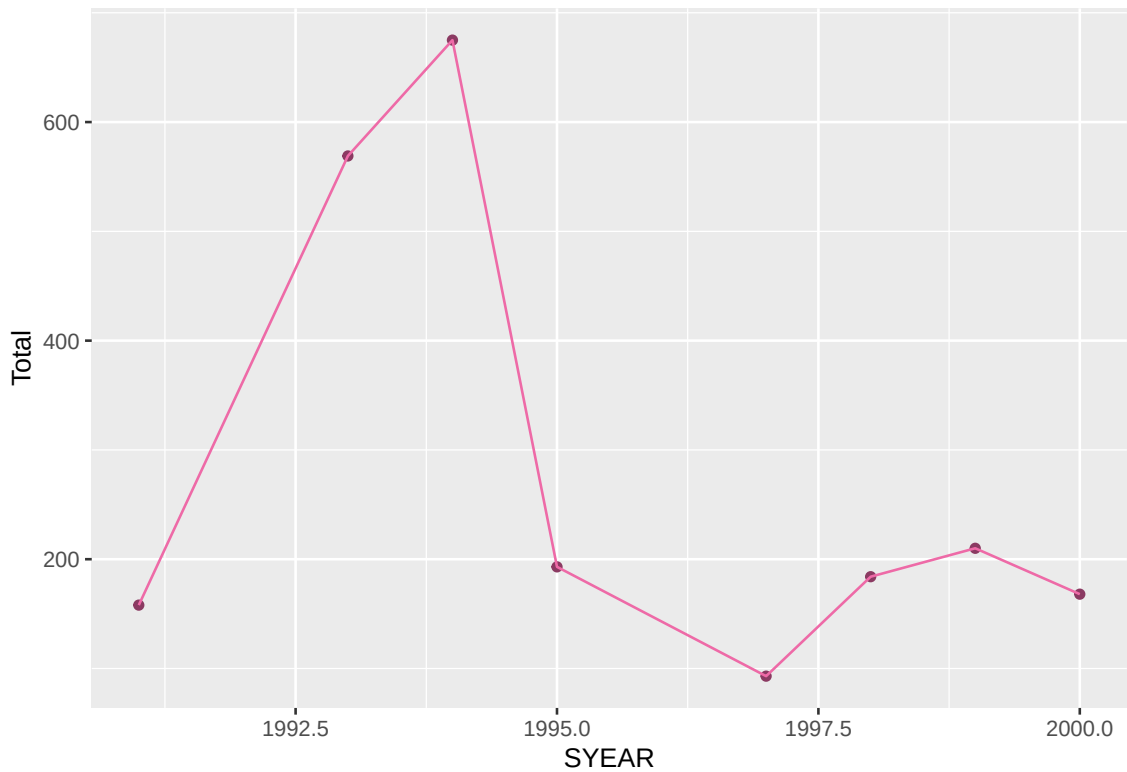
ggplot(baseline3,aes(Total))+
  geom_histogram(col="hotpink4",fill="hotpink2")

## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
#variation over time
baseline=Baseline%>%
  group_by(SYEAR)%>%
  summarise(Total=(n_distinct(VALUE[FIELDNAME=="VEG_SPEC"] ,na.rm=TRUE)))

ggplot(baseline,aes(SYEAR,Total))+
  geom_point(col="hotpink4")+
  geom_line(col="hotpink2")
```

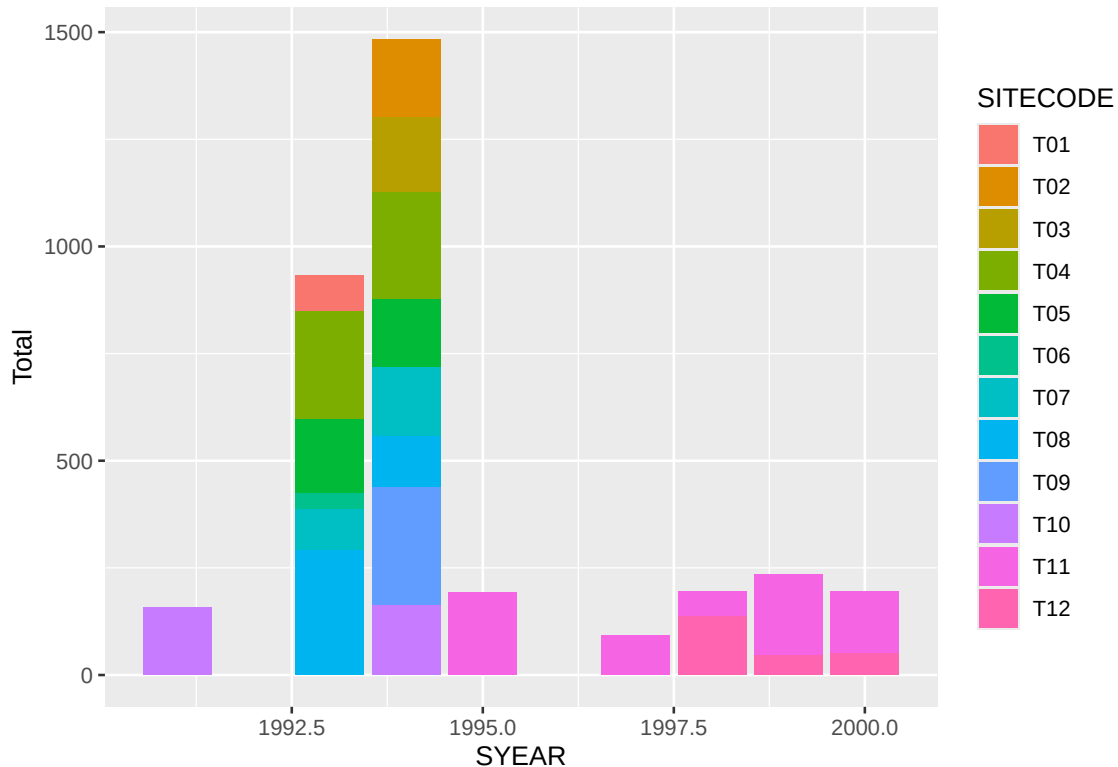


```
#spatial variation
```

```
baseline = Baseline%>%
  group_by(SYEAR,SITECODE)%>%
  summarise(Total=(n_distinct(VALUE[FIELDNAME=="VEG_SPEC"])),na.rm=TRUE)

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by SYEAR and SITECODE.
## i Output is grouped by SYEAR.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(SYEAR, SITECODE))` for per-operation grouping
## (`?dplyr::dplyr_by`) instead.

#stacked bar chart
ggplot(baseline,aes(SYEAR>Total,fill=SITECODE))+
  geom_bar(stat="identity")
```



a. The chosen variable was number of different things counted as part of each sample, this goes between 0 and 50. The data is slightly positively skewed with a mode at 11 and a few outliers towards the top end. There are no more zeros than you would expect, for some reason the data seems to have been collected differently than finegrain and woodland so any samples that spotted nothing were included this time.

b. We investigate total number of species found each year, I chose a line graph to show how it changes over time, the plot shows that findings were inconsistent each year this may be because of less samples being taken or something happening in the area like flooding that caused it to significantly decrease after 1994.

c. We investigate number of species found in each site each year, I use a stacked bar chart to show how totals change over time as well as totals for specific sites. The plot shows that there were only two years where most sites were sampled which makes sense given that it is a baseline but strangely T11 and T12 were sampled over several years, this may be that different plots were sampled at different times or they misunderstood the instructions or all of the other sites were not sampled when they should have been.

Butterflies

We exploited the butterfly count data by looking at how the values are distributed, how they change over time, how they vary between sites, and whether there's any clear link between counts and year. We used a

Most of the counts are very low, often zero or just one, with only a few higher values reaching up to 10. The average (around 2.36) is higher than the median (1), which shows that a small number of larger counts are pulling the average up.

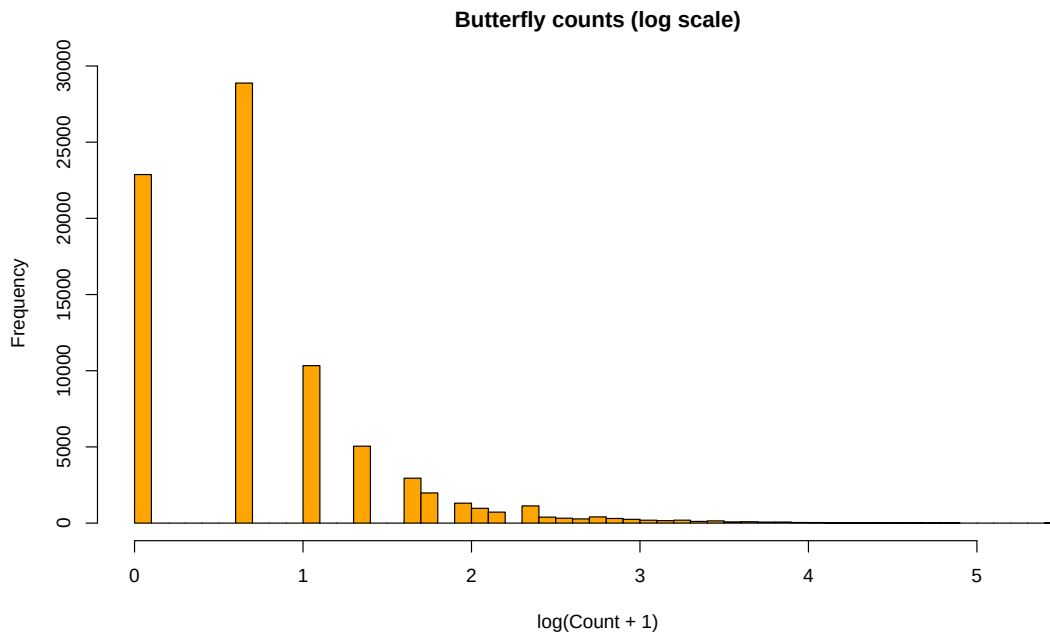
Looking at the data over time, the number of observations changes quite a bit. There are more records in the mid-1990s and again in the mid-2000s, but the number drops off after 2005. This suggests that the data reflects changes in how often surveys were done, rather than a clear trend in butterfly counts.

There are also noticeable differences between sites. Some, like T09 and T10, tend to have higher counts than others. This points to differences in local conditions or possibly how surveys were carried out at each site. Overall, there's no clear pattern. High and low values appear throughout the timeline without a steady increase or decrease.

Overall, the dataset is quite large and useful, but the high number of zeros, the skewed distribution of counts, and the irregular sampling over time are things to keep in mind when analyzing the data.

```
## R code to clean the data and create a histogram
r
bf <- Butterflies
bf_clean <- bf[!is.na(bf$VALUE), ]
bf_clean$SDATE <- as.Date(bf_clean$SDATE, format="%d-%b-%y")
bf_clean$YEAR <- as.numeric(format(bf_clean$SDATE, "%Y"))

hist(log1p(bf_clean$VALUE),
     breaks=50,
     col="orange",
     main="Butterfly counts (log scale)",
     xlab="log(Count + 1)",
     ylab="Frequency")
```



```
## R code to create a bar plot of butterfly sampling over time
bf_year <- table(bf_clean$YEAR)

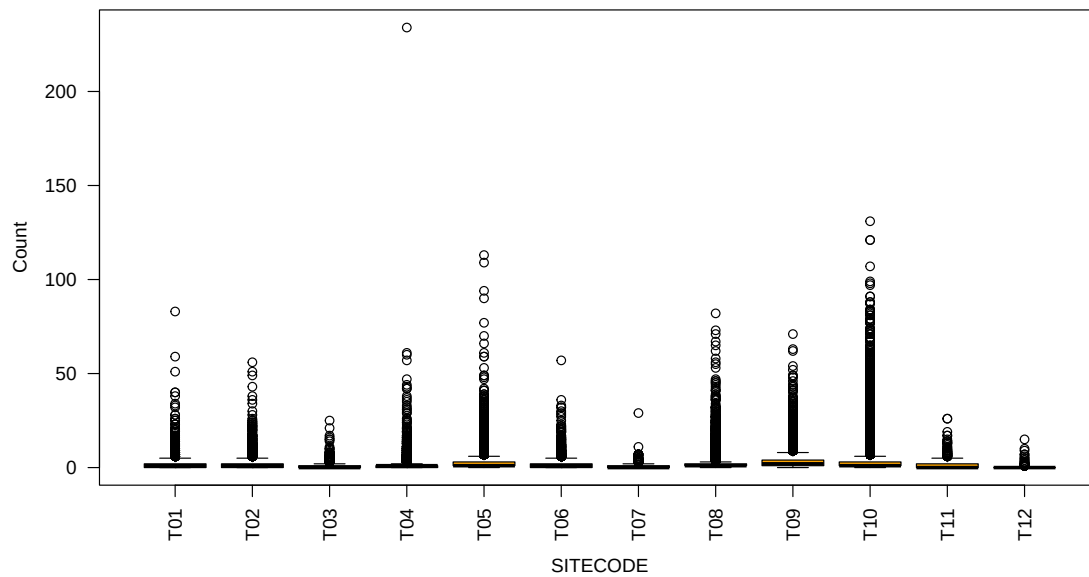
plot(as.numeric(names(bf_year)), bf_year,
     type="l", col="orange", lwd=2,
     main="Butterfly sampling over time",
     xlab="Year", ylab="Observations")
```

Butterfly sampling over time

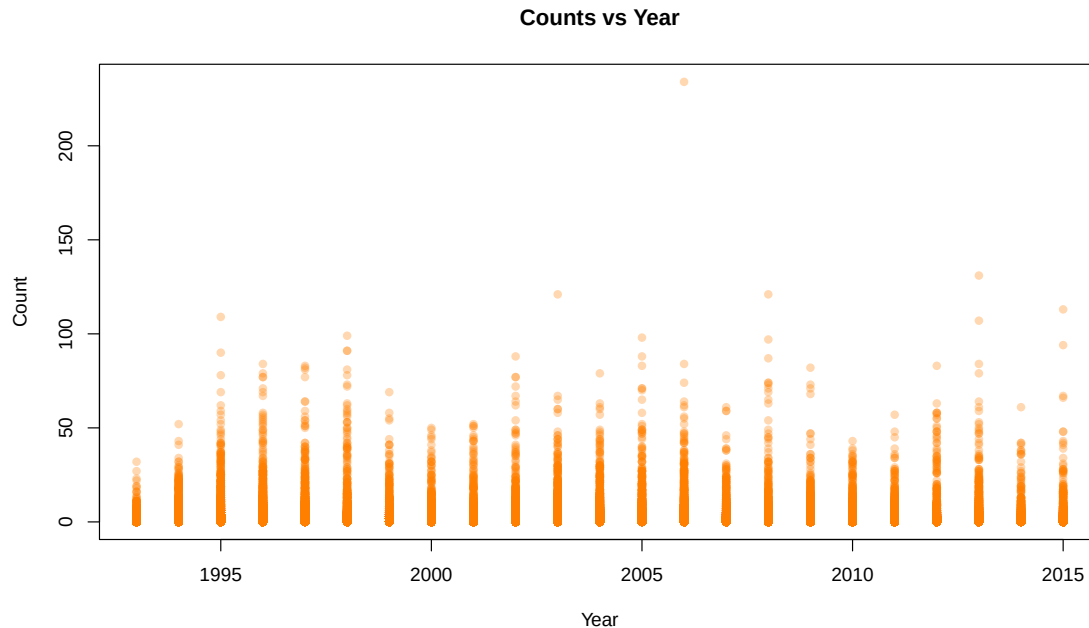


```
boxplot(VALUE ~ SITECODE,
  data=bf_clean,
  las=2,
  col="orange",
  main="Butterfly counts by site",
  ylab="Count")
```

Butterfly counts by site



```
plot(bf_clean$YEAR, bf_clean$VALUE,
  col=rgb(1,0.5,0,0.3),
  pch=16,
  main="Counts vs Year",
  xlab="Year", ylab="Count")
```



```
summary(bf_clean$VALUE)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.000   0.000    1.000   2.359   2.000  234.000
```

Moths

We looked at the moth count data in a few different ways: by checking how the values are spread out, how they change over time, how they differ between sites, and how counts relate to the year. This included things like a log-scale histogram, a time series, boxplots for each site, and a scatterplot of counts versus year. These help us understand the overall shape of the data, how often we recorded moths over time, differences between locations, and whether there are any trends.

The counts are very uneven. Most of the values are really low, often zero or just one, but there are a few very large numbers (up to 1666). Even after using a log scale, the data is heavily skewed. The average count (about 4.15) is much higher than the middle value (median = 1), which shows that a few big counts are pulling the average higher up. This kind of pattern is common in insect trap data.

When we look at the data over time, the number of observations changes a lot. There was more sampling in the mid-1990s and early 2000s, but it drops off in later years. This likely reflects changes in how often trapping was done, rather than a clear change in moth populations.

There are also clear differences between sites. Some locations, especially T06 and T12, tend to have higher counts and more variation, while most other sites mostly have low counts. This suggests that local conditions at each site play a big role.

Finally, when comparing counts to year, there isn't a clear pattern. High values appear randomly rather than following a steady increase or decrease. Overall, the dataset gives useful information, but it's important to keep in mind the large number of zeros, the occasional extreme values, and the uneven sampling over time when interpreting the results.

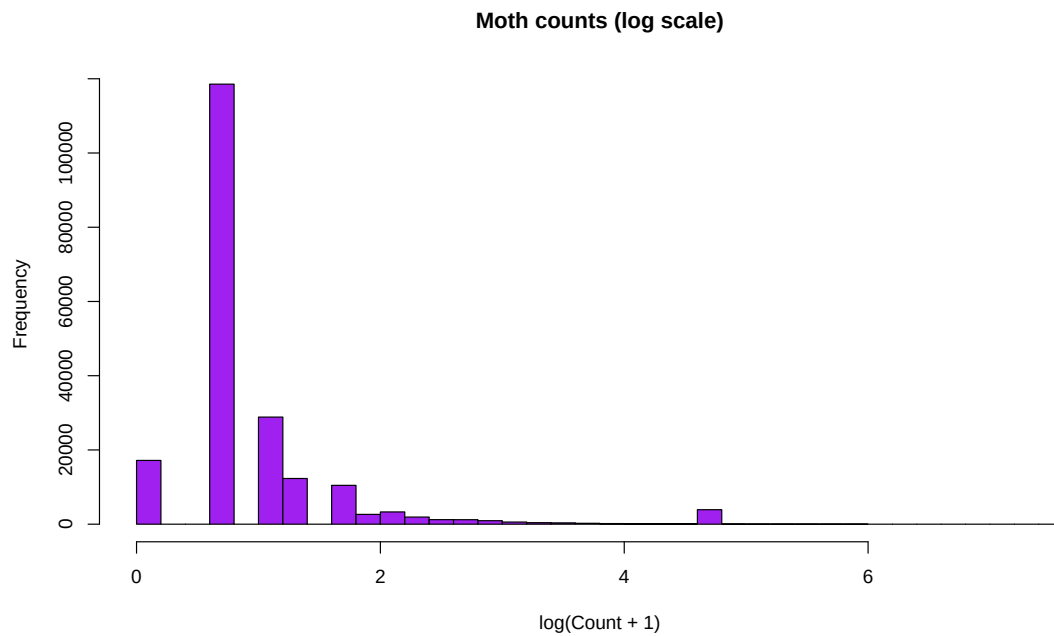
```
mt <- Moths

mt$VALUE_num <- suppressWarnings(as.numeric(mt$VALUE))
mt_clean <- mt[!is.na(mt$VALUE_num), ]
mt_clean$SDATE <- as.Date(mt_clean$SDATE, format="%d-%b-%y")
mt_clean$YEAR <- as.numeric(format(mt_clean$SDATE, "%Y"))
mt_clean <- mt_clean[!is.na(mt_clean$YEAR), ]

hist(log1p(mt_clean$VALUE_num),
     breaks=50,
     col="purple",
```

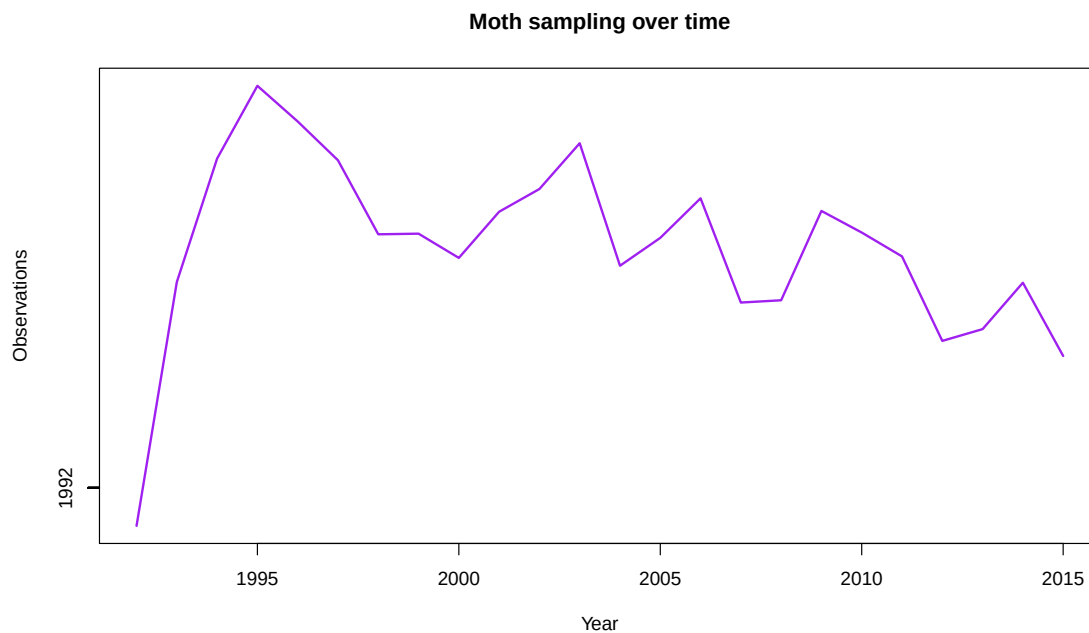


```
main="Moth counts (log scale)",
xlab="log(Count + 1)",
ylab="Frequency")
```

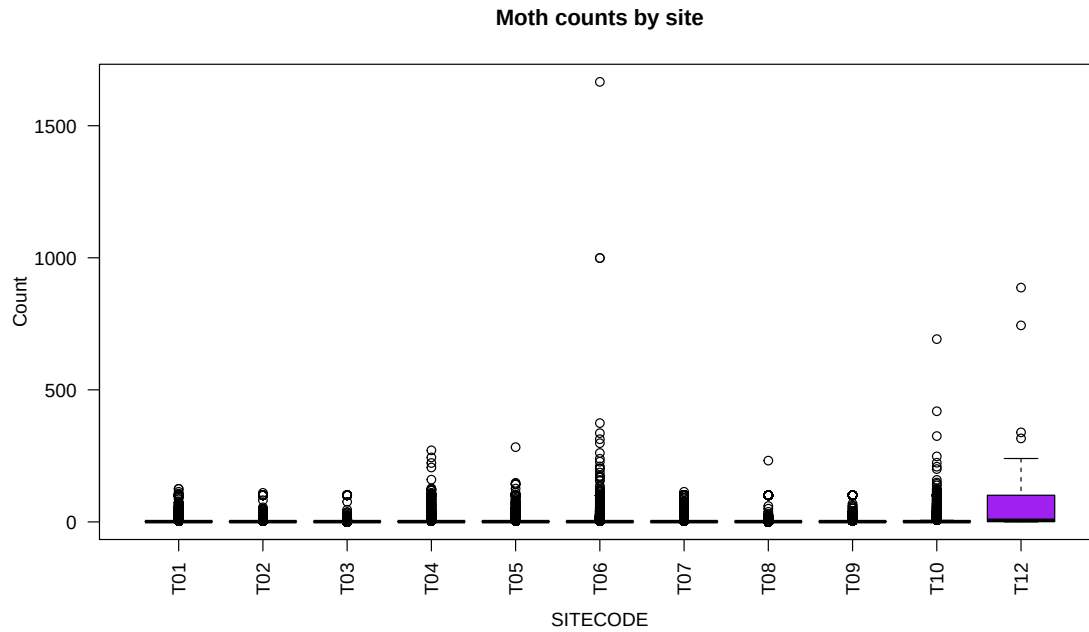


```
mt_year <- table(mt_clean$YEAR)

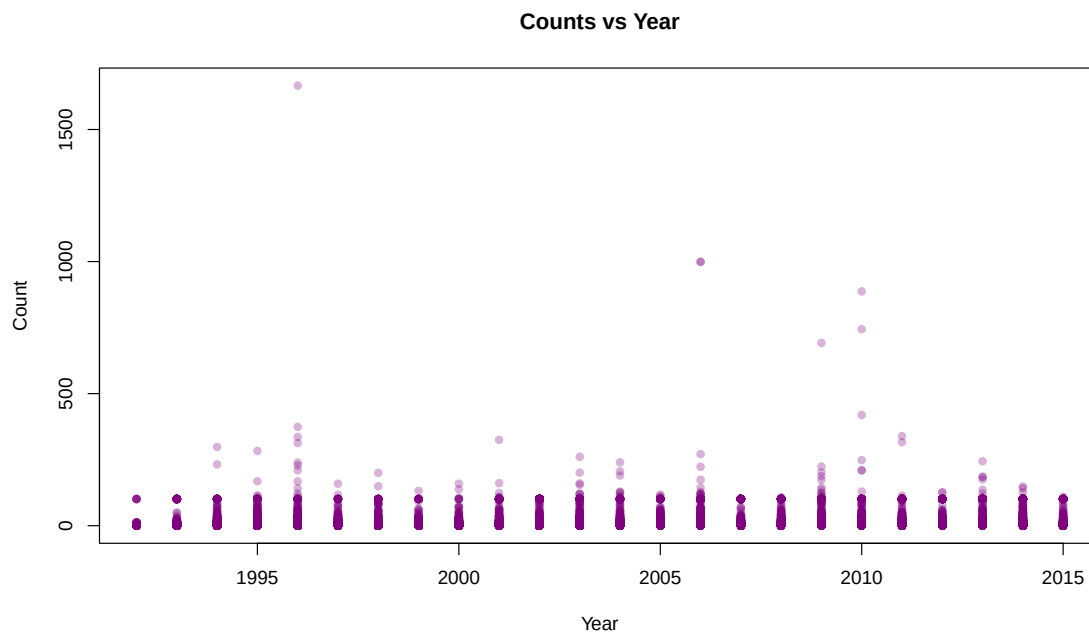
plot(as.numeric(names(mt_year)), mt_year,
     type="l", col="purple", lwd=2,
     main="Moth sampling over time",
     xlab="Year", ylab="Observations")
```



```
boxplot(VALUE_num ~ SITECODE,
        data=mt_clean,
        las=2,
        col="purple",
        main="Moth counts by site",
        ylab="Count")
```



```
plot(mt_clean$YEAR, mt_clean$VALUE_num,
     col=rgb(0.5,0,0.5,0.3),
     pch=16,
     main="Counts vs Year",
     xlab="Year", ylab="Count")
```



```
summary(mt_clean$VALUE_num)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
##    0.000    1.000    1.000    4.149    2.000 1666.000
```

Task 5: Cross-data comparison for vegetation

###Question: This comparison examines whether differences between a woodland-focused subset of VF and VW mainly reflect ecological pattern or sampling design.

Data preparation

```
vf1 <- FineGrain
vf2 <- FineGrain2
vw1 <- Woodland
```

Construct comparable subsets

```
shared_sites <- intersect(unique(vf1$SITECODE), unique(vw1$SITECODE))

vf2_sub <- vf2[
  vf2$SITECODE %in% shared_sites &
  !is.na(vf2$LANDUSE) &
  vf2$LANDUSE %in% c(11, 12, 13),
  c("SITECODE", "SYEAR", "PLOTPID", "LANDUSE")
]

vf_join <- merge(
  vf1,
  vf2_sub,
  by = c("SITECODE", "SYEAR", "PLOTPID")
)

vf_tmp <- vf_join[
  vf_join$FIELDNAME == "VEG_SPEC" &
  !is.na(vf_join$VALUE) &
  vf_join$VALUE > 0,
  c("SITECODE", "SYEAR", "PLOTPID", "VALUE")
]

vf_rich <- aggregate(
  VALUE ~ SITECODE + SYEAR + PLOTPID,
  data = vf_tmp,
  FUN = function(x) length(unique(x))
)
names(vf_rich)[names(vf_rich) == "VALUE"] <- "richness"
vf_rich$dataset <- "VF woodland subset"

vw_tmp <- vw1[
  vw1$FIELDNAME == "SEEDLING" &
  !is.na(vw1$VALUE) &
  vw1$VALUE > 0,
  c("SITECODE", "SYEAR", "PLOTPID", "VALUE")
]

vw_rich <- aggregate(
  VALUE ~ SITECODE + SYEAR + PLOTPID,
  data = vw_tmp,
  FUN = function(x) length(unique(x))
)
names(vw_rich)[names(vw_rich) == "VALUE"] <- "richness"
vw_rich$dataset <- "VW seedling richness"

veg_compare <- rbind(
  vf_rich[, c("SITECODE", "SYEAR", "PLOTPID", "richness", "dataset")],
  vw_rich[, c("SITECODE", "SYEAR", "PLOTPID", "richness", "dataset")]
)

head(veg_compare)

##   SITECODE SYEAR PLOTPID richness      dataset
```

```
## 1      T06  2013      13      13 VF woodland subset
## 2      T06  2013      14       9 VF woodland subset
## 3      T06  2013      15      11 VF woodland subset
## 4      T06  2013      16      10 VF woodland subset
## 5      T06  2013      17       9 VF woodland subset
## 6      T06  2013      18      14 VF woodland subset
```

Summary table

```
summary_fun <- function(x) {
  c(
    n = length(x),
    median_richness = median(x),
    q1 = as.numeric(quantile(x, 0.25)),
    q3 = as.numeric(quantile(x, 0.75))
  )
}

veg_table <- as.data.frame(do.call(
  rbind,
  tapply(veg_compare$richness, veg_compare$dataset, summary_fun, simplify = FALSE)
))

veg_table$dataset <- rownames(veg_table)
rownames(veg_table) <- NULL
veg_table <- veg_table[, c("dataset", "n", "median_richness", "q1", "q3")]

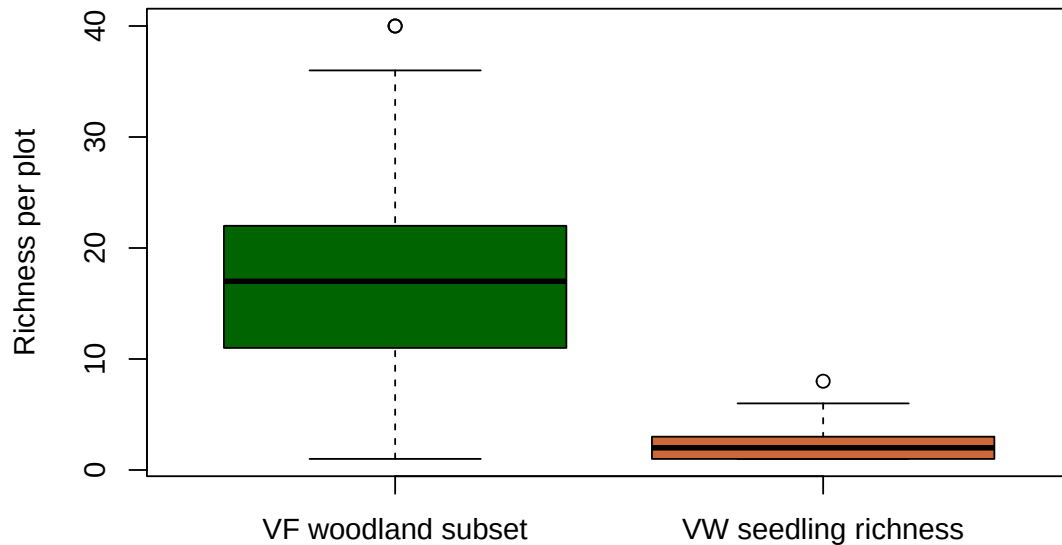
veg_table
```

```
##              dataset    n median_richness q1 q3
## 1  VF woodland subset 301             17 11 22
## 2  VW seedling richness 269              2  1  3
```

Figure

```
boxplot(
  richness ~ dataset,
  data = veg_compare,
  col = c("darkgreen", "sienna3"),
  main = "Task 5. Woodland-related richness comparison between VF and VW",
  xlab = "",
  ylab = "Richness per plot"
)
```

Task 5. Woodland-related richness comparison between VF and VW



Justification:

To make the datasets comparable, the VF subset was restricted to plot-years classified as woodland, coniferous woodland or scrub at the four sites shared with VW. The VF summary metric is plot richness across the ten sampled cells, while the VW metric is seedling-species richness per plot, which is the closest directly comparable richness measure available in the woodland file.

Results:

VF woodland/scrub plots are much richer than VW plots. The VF subset has a median richness of 17 species whereas VW has a median seedling richness of 1 species. The two distributions overlap only slightly.

Interpretation:

The difference is driven mainly by sampling design rather than by ecological contrast alone. VF records all taxa present in sampled ground-layer cells, including non-woody species, bryophytes and lichens, so richness is necessarily much higher. By contrast, VW is designed to capture woodland structure and regeneration, so seedling richness represents a much narrower component of vegetation change.

Task 6: Cross-data comparison for invertebrates

Limits: 200 words text + 1 figure + 1 table.

Question

This section compares the butterfly dataset (IB) and the moth dataset (IM) to assess whether the observed difference mainly reflects ecological differences between the groups or differences in monitoring design.

Data preparation

```
ib <- Butterflies
im <- Moths
```

```

ib$VALUE <- suppressWarnings(as.numeric(ib$VALUE))
im$VALUE <- suppressWarnings(as.numeric(im$VALUE))

ib$DATE <- as.Date(ib$SDATE, format = "%d-%b-%y")
im$DATE <- as.Date(im$SDATE, format = "%d-%b-%y")

ib$YEAR <- as.numeric(format(ib$DATE, "%Y"))
im$YEAR <- as.numeric(format(im$DATE, "%Y"))
ib$MONTH <- as.numeric(format(ib$DATE, "%m"))
im$MONTH <- as.numeric(format(im$DATE, "%m"))

```

Construct comparable subsets

```

shared_sites <- intersect(unique(ib$SITECODE), unique(im$SITECODE))

# -----
# Butterflies (IB)
# -----
ib_sub <- ib[
  ib$SITECODE %in% shared_sites &
  !is.na(ib$VALUE) &
  !is.na(ib$MONTH) &
  ib$MONTH %in% 4:9 &
  ib$FIELDNAME != "XX",
]

ib_abund <- aggregate(
  VALUE ~ SITECODE + SDATE + MONTH,
  data = ib_sub,
  FUN = sum
)
names(ib_abund)[names(ib_abund) == "VALUE"] <- "abundance"

ib_rich <- aggregate(
  FIELDNAME ~ SITECODE + SDATE + MONTH,
  data = ib_sub[ib_sub$VALUE > 0, ],
  FUN = function(x) length(unique(x))
)
names(ib_rich)[names(ib_rich) == "FIELDNAME"] <- "richness"

ib_daily <- merge(ib_abund, ib_rich, by = c("SITECODE", "SDATE", "MONTH"), all = TRUE)
ib_daily$richness[is.na(ib_daily$richness)] <- 0
ib_daily$dataset <- "Butterflies"

# -----
# Moths (IM)
# -----
im_sub <- im[
  im$SITECODE %in% shared_sites &
  !is.na(im$VALUE) &
  !is.na(im$MONTH) &
  im$MONTH %in% 4:9 &
  !grepl("^Q", im$FIELDNAME) &
  im$FIELDNAME != "XX",
]

im_abund <- aggregate(
  VALUE ~ SITECODE + SDATE + MONTH,
  data = im_sub,
  FUN = sum
)

```

```

)
names(im_abund)[names(im_abund) == "VALUE"] <- "abundance"

im_rich <- aggregate(
  FIELDNAME ~ SITECODE + SDATE + MONTH,
  data = im_sub[im_sub$VALUE > 0, ],
  FUN = function(x) length(unique(x))
)
names(im_rich)[names(im_rich) == "FIELDNAME"] <- "richness"

im_daily <- merge(im_abund, im_rich, by = c("SITECODE", "SDATE", "MONTH"), all = TRUE)
im_daily$richness[is.na(im_daily$richness)] <- 0
im_daily$dataset <- "Moths"

inv_compare <- rbind(
  ib_daily[, c("SITECODE", "SDATE", "MONTH", "abundance", "richness", "dataset")],
  im_daily[, c("SITECODE", "SDATE", "MONTH", "abundance", "richness", "dataset")]
)

head(inv_compare)

```

```

##   SITECODE   SDATE MONTH abundance richness   dataset
## 1      T01 01-APR-97     4         16         2 Butterflies
## 2      T01 01-APR-99     4          7         2 Butterflies
## 3      T01 01-AUG-00     8        104        11 Butterflies
## 4      T01 01-AUG-01     8         56         7 Butterflies
## 5      T01 01-AUG-97     8        140         6 Butterflies
## 6      T01 01-JUL-09     7        105        10 Butterflies

```

Summary table

```

summary_fun <- function(a, r) {
  c(
    n = length(a),
    median_abundance = median(a),
    median_richness = median(r)
  )
}

ib_summary <- summary_fun(ib_daily$abundance, ib_daily$richness)
im_summary <- summary_fun(im_daily$abundance, im_daily$richness)

inv_table <- data.frame(
  dataset = c("Butterflies", "Moths"),
  n = c(ib_summary["n"], im_summary["n"]),
  median_abundance = c(ib_summary["median_abundance"], im_summary["median_abundance"]),
  median_richness = c(ib_summary["median_richness"], im_summary["median_richness"])
)

inv_table

```

```

##      dataset      n median_abundance median_richness
## 1 Butterflies 3985          16          4
## 2      Moths 26163           7          4

```

Seasonal profile figure

```

season_compare <- aggregate(
  abundance ~ dataset + MONTH,
  data = inv_compare,
  FUN = median
)

```

```

)
names(season_compare)[names(season_compare) == "abundance"] <- "median_abundance"

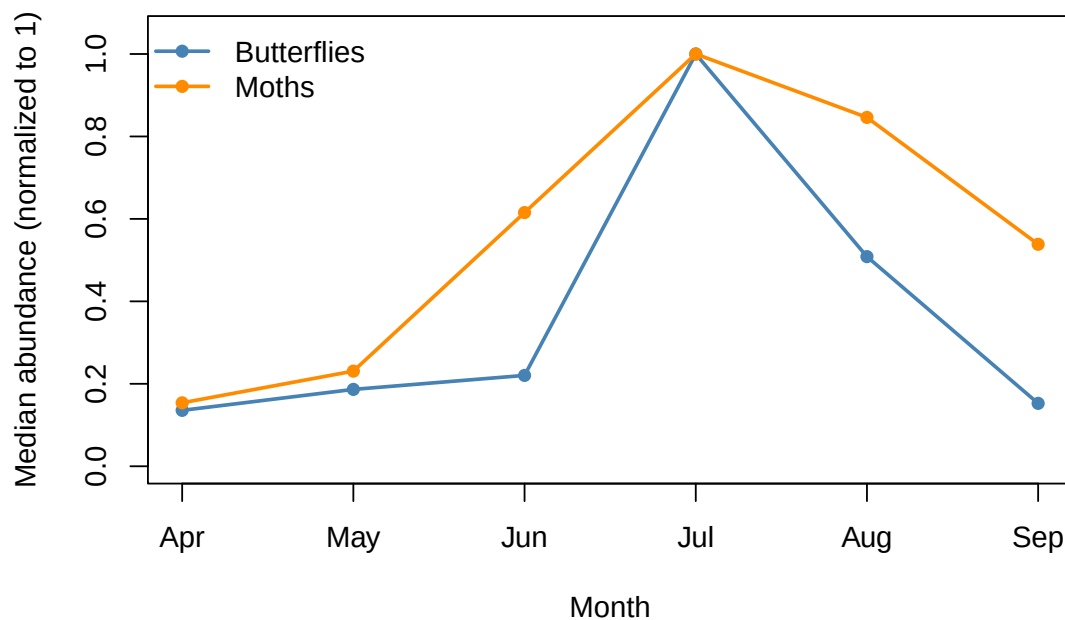
season_compare$normalized <- ave(
  season_compare$median_abundance,
  season_compare$dataset,
  FUN = function(x) x / max(x)
)

bf <- season_compare[season_compare$dataset == "Butterflies", ]
mt <- season_compare[season_compare$dataset == "Moths", ]

plot(
  bf$MONTH, bf$normalized,
  type = "o", pch = 16, lwd = 2, col = "steelblue",
  ylim = c(0, 1.05), xaxt = "n",
  xlab = "Month",
  ylab = "Median abundance (normalized to 1)",
  main = "Task 6. Normalized seasonal profiles for butterflies and moths"
)
axis(1, at = 4:9, labels = c("Apr", "May", "Jun", "Jul", "Aug", "Sep"))
lines(mt$MONTH, mt$normalized, type = "o", pch = 16, lwd = 2, col = "darkorange")
legend(
  "topleft",
  legend = c("Butterflies", "Moths"),
  col = c("steelblue", "darkorange"),
  lwd = 2,
  pch = 16,
  bty = "n"
)

```

Task 6. Normalized seasonal profiles for butterflies and moths



Justification

The comparison was restricted to sites common to both datasets. IB was summarised at the survey-day level and IM at the trap-day level. IM was limited to April–September to match the

butterfly monitoring window as closely as possible.

Results

In this version of the analysis, both datasets have the same median sample-level richness of 4 species. Median abundance is 16 individuals per sample for butterflies and 7 for moths. Both seasonal profiles peak in July, but moth abundance remains relatively high into August and September, while butterfly abundance declines more sharply after July.

Interpretation

The shared July peak suggests a genuine seasonal signal, but differences in the profile shape also reflect method. IB is a daytime visual transect survey, whereas IM is a trap-based nocturnal sampling design.